

RESEARCH MONOGRAPH

Novelty That Resembles the Past

Can language models recognize
what they have never encountered?

Foundations, informational limits
and experimental validation

An analysis of the difference between genuine novelty,
familiarity and the appearance of originality.

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Preface. The Problem of an Evaluator Educated by the Past

A system is asked to read a thousand proposals and retain ten genuinely new ideas. It was trained on texts from the past, learned regularities of scientific discourse and recognizes expressions such as “a new perspective,” “a unified framework” or “an unexplored combination.” The uncomfortable question is not whether the system can produce ten elegant justifications. The question is whether what it rejects includes precisely the contribution that does not yet sufficiently resemble what the community calls innovation.

This book investigates a precise suspicion: language models might confuse novelty with the familiarity of patterns of novelty. The hypothesis is neither ridiculed through an abstract appeal to “generalization” nor turned into a theorem merely because it sounds plausible. It is broken down into different claims with different truth conditions. Some describe experimental risks. Others are demonstrable informational limits. Still others are too strong and admit constructive counterexamples.

The position defended here is that a model trained on the past is not, for that reason alone, incapable of recognizing a new difference. Conversely, a system that has lost the relevant difference before evaluation can no longer recover it from the remaining representation unless it receives additional information. Almost the entire practical problem lies between these two claims. We must determine what information the evaluator sees, what comparison it performs, what it understands by prior work and what evidence allows it to move from resemblance to identity or from difference to contribution.

The text is intended for researchers and programmers who want to design tools for analyzing ideas, selecting proposals, examining the literature or automating research. The intended level is advanced, but the necessary concepts are introduced before they are used. Familiarity with information theory, causal inference or computability is not

assumed. What is assumed is the reader's willingness to follow arguments, distinguish definitions from discoveries and accept that a rigorous negative result can be more useful than an impressive score.

One editorial distinction is essential. Accounts of published experiments are accompanied by numbered references. Working definitions, elementary proofs, constructed examples and proposed protocols are presented as such. The book does not report a new benchmark run on commercial or open models. The small calculations and logical examples are didactic checks; they are not presented as empirical findings about an LLM's capabilities. The architectural proposals are research projects, not descriptions of an already validated product.

The literature is a critical selection of primary sources, checked through September 11, 2026, not an exhaustive systematic review. The versions consulted are identified in the bibliography. Results concerning particular models, datasets and procedures are not automatically generalized to all future models. Likewise, agreement with a reviewer is not treated as identity with historical truth, and failure to find prior work in a search engine's results is not treated as proof of its absence.

The book can be read sequentially, from foundations toward experiment, or used as a design manual. The chapters on information and computability establish what can coherently be required of an evaluator. The methodological chapters show how a failure mechanism can be identified. The final chapters discuss how to build a system that preserves its ability to be corrected. The appendices offer additional proofs, a preregistration protocol, an executable example and a technical vocabulary.

The aim is neither to defend nor condemn a class of models. It is to protect a fragile intellectual operation: noticing a difference before the language, statistics, criteria or institution evaluating it reduces it to something already known.

Chapter 1. What the Hypothesis Actually Claims

1.1. From Intuitive Suspicion to Separable Propositions

The formula “an LLM finds only what it has encountered before” mixes at least two senses of the verb to find. A system can retrieve a stored example, but it can also apply a learned rule to an instance that did not exist in training. To recognize that two numbers differ, it need not have previously memorized their pair. To determine that a program produces an output absent from a finite library, it need not have encountered that particular program. The fact that a competence was learned from the past does not imply that the objects on which it can operate are limited to the past.

This observation does not solve the problem of language models. It eliminates only an invalid inference. It remains to be established whether the model actually applies a comparison procedure or responds on the basis of superficial regularities. An architecture might allow an operation without the training procedure inducing it sufficiently well. A competence might exist on simple problems and be lost on long texts. A difference might be recognized correctly but wrongly interpreted as unimportant. Logical possibility and actual performance must therefore be examined separately.

The behavioral form of the hypothesis says that the evaluator favors ideas expressed through familiar patterns of innovation even when the semantic contribution is held constant. The causal form claims that this preference is produced by frequencies or correlations in training. The informational form claims that certain intermediate representations erase the differences needed for classification. The universal form claims that no LLM can detect a genuinely new contribution. These propositions are not equivalent. A proof of the informational form for a particular representation does not prove the universal form. A correlation between perplexity and score does not, by itself, identify the training process as the cause.

There is also an institutional form of the suspicion: selection systems can turn a modest preference into an important cumulative effect. If a filter more often rejects ideas with unusual vocabulary, and only accepted ideas are then discussed, cited and used to train other filters, the space of future examples may narrow. This is a hypothesis about a dynamic process, not a property demonstrated by merely analyzing a single answer. It requires observations of selection, allocated resources and information fed back into the system.

To avoid ambiguity, we will call familiarity the fit with regularities already learned or frequent in a representation; we will call novelty the relevant difference relative to an explicit reference corpus. Familiarity can be linguistic, conceptual, methodological or social. Novelty can concern the mechanism, result, applicability conditions or proof. A highly familiar formulation can describe a new contribution. A strange formulation can conceal a banal idea. Neither “familiar” nor “new” should be used without specifying the object of comparison.

A constructed example shows why this discipline matters. The first proposal says that a protocol works when fewer than one third of the participants behave adversarially. The second claims the same guarantee under a different assumption about synchrony. A thematic summary can represent both texts as “a distributed protocol tolerant of malicious participants.” At this level they seem identical. At the level of assumptions and guarantees, decisive differences may exist. It is not necessary to assume that the second proposal is correct. To detect the claimed contribution, it is enough to notice that the claims are not the same.

The example introduces a separation that will recur throughout the book. Identifying the stated difference, checking whether prior work exists for that difference and demonstrating its validity are distinct operations. If an evaluator compresses them into a single verdict, an error can no longer be diagnosed. “It is not new” may mean that the mechanism is old, that the model has not understood it, that it does not believe the proof or that it prefers a more popular problem. A score without decomposition can conceal all these situations.

A good hypothesis must also specify what result would weaken it. If a model receives two contributions new to it, the correct prior work and an exact criterion, then robustly identifies the difference under

controlled reformulations, the claim of general impossibility becomes untenable. If, however, novelty is detected only when the text contains conventional expressions of originality, the behavioral suspicion gains strength. If changing training frequencies changes evaluations without changing the ground truth, a causal explanation also comes closer.

The book's methodological position can be stated simply: it is not enough to ask whether the model "understands novelty." We must construct situations in which an explanation based on understanding the difference and one based on recognizing the packaging make different predictions. Only then does the experiment separate the mechanisms.

1.2. Novelty Is Not Rarity, Truth or Value

Rarity is a property of a distribution. Novelty is a relation to prior work. An old method ignored by the current community may be rare in the analyzed corpus yet not historically new. A recent discovery may be repeated in thousands of documents and still remain new relative to the state of knowledge last year. Corpus frequency, training exposure and historical priority are different variables. Confusing them can produce both enthusiasm for rediscoveries and rejection of discoveries that have already become visible.

Linguistic novelty concerns the form of expression. Conceptual novelty concerns what is claimed or the mechanism proposed. Operational novelty concerns a new capacity to do something. Epistemic novelty concerns what becomes known or justified for an agent or community. Historical novelty concerns the absence of relevant prior work before a given time. These distinctions are the book's working definitions. They do not claim that a single taxonomy accepted by all disciplines exists.

A new proof of a known theorem illustrates the independence of the dimensions. The conclusion may be old, the proof method new and verification time shorter. An algorithm may compute the same function but have different complexity, memory requirements or privacy properties. If the evaluator reduces the contribution to the input-output relation, it will legitimately declare the absence of functional nov-

elty, but will be wrong if it extends the conclusion to algorithmic novelty in general. The semantic object chosen determines what can be observed.

Truth constitutes another axis. A hypothesis can be original and false. An observation can be true and well known. A proposal can be original in its precise form but impossible under its stated conditions. In a research tool, identifying a claimed contribution need not be suspended until its complete validation. It is useful to know exactly what the author claims to change before checking whether the change is feasible. At the same time, the originality of a false claim must not automatically be converted into scientific merit.

Value depends on purposes and consequences. A minor variation can have great practical value if it eliminates a dominant cost in an application. A highly original construction may currently have limited usefulness. These judgments can be perfectly compatible. The problem arises when the evaluator calls everything it likes “novelty” or everything it finds useless “lack of novelty.” At that point, the criterion becomes a global preference and conceptual precision disappears.

Originality also has a provenance dimension. Two people can independently produce the same idea. The second person’s contribution may be psychologically original without having historical priority. Conversely, a text may present an idea unknown to the evaluator but borrowed from an omitted source. Detecting prior work does not automatically establish the author’s intention. An accusation of plagiarism involves questions of access, attribution and source-use practices that cannot be reduced to the semantic identity of two texts.

Boden distinguishes combinatorial, exploratory and transformational forms of creativity. This perspective is useful because changing the rules of the search space is not the same as finding another solution within existing rules. However, the typology of creativity does not by itself provide a detector of historical priority. It helps formulate the type of change we should look for. [R01]

We can therefore define a contribution profile instead of a single score. The profile can separate the result, mechanism, assumptions, justification procedure and scope of application. Not every dimension must be numerical. Sometimes a comparative statement is more in-

formative than a number: “the same guarantee, but without the independence assumption” or “the same function, but with memory sub-linear in the stated parameter.” Such formulations force the evaluator to specify the difference it recognizes.

1.3. Reference, Criterion and Time Are Part of the Problem

We will denote a novelty judgment by the following expression, in which every argument matters:

$$N(x;K_t,C,B).$$

Here, x is the contribution being evaluated, K_t is the body of knowledge available up to time t , C represents the criterion of equivalence or relevance, and B describes the budget for access, computation and verification. We are not proposing a magic formula through this notation. It serves to prevent omissions. A novelty claim without a reference corpus, a criterion and declared resources is generally more ambiguous than it appears.

The criterion C may say that two contributions are equivalent if they have the same function on the same domain. In another application, equivalence may require the same security assumptions, the same complexity and the same adversarial model. A change that disappears under the first criterion may remain essential under the second. In this sense, novelty is not arbitrary, but it depends on the question asked. Once the criterion is fixed, the comparison can become very precise.

The corpus K_t must be distinguished from the whole world. The indexed library, the documents accessible to the evaluator and all existing prior work are not the same set. An unindexed article, a thesis in another language or an earlier version of a report can change the historical verdict. The system should not be condemned for lacking inaccessible information. It must, however, be prevented from turning limited access into unlimited certainty.

Time introduces a further difficulty. When we evaluate an idea retrospectively, the model may know later versions, replications, controversies or the community's assessment. Even if we remove the authors' names, information about the result may survive in the wording. A test asking "how new was this contribution in year t ?" is not automatically valid if the model was trained on documents after t . It may recognize the discovery's reputation instead of reconstructing the situation at the time.

The budget B is included because being able to verify in principle does not mean being able to verify with the available resources. Prior work may be retrievable after a two-day search, but not within the thirty seconds allotted to filtering. An equivalence may be decidable but too costly for a million candidates. Comparing systems without equalizing the budget risks turning an access or computational advantage into a supposed conceptual superiority.

One might object that adding these parameters dilutes the notion of novelty. The objection is legitimate if every failure is excused by conveniently changing the definition. That is why the parameters must be fixed before evaluation. Relativization serves not to avoid testing but to make the verdict testable. "New relative to library K , under criterion C , with the decisive prior work supplied" is a narrower claim but methodologically stronger than "the model understands originality."

A practical detector can produce three types of conclusions. It can identify prior work that anticipates the contribution. It can identify a difference relative to the prior work examined. It can determine that the data are insufficient for a conclusion. These should not be artificially arranged on a single ordinal scale. Missing information is not medium novelty, and the existence of an unvalidated difference is not evidence of importance.

At the end of this clarification, the initial hypothesis becomes more productive. The question is no longer whether the past forbids the future. It is whether the procedure learned from the past preserves and examines those differences through which the future might be more than a reformulation of the past. This question allows both proofs of conditional impossibility and constructions that demonstrate possibility.

Chapter 2. Probability, Information and Surprise

2.1. Why Low Probability Does Not Mean High Novelty

A probabilistic model assigns weights to possible outcomes. If we have a distribution p over a vocabulary, $p(w)$ describes the probability of a symbol w under the modeled conditions. In a language model, the distribution is conditioned on context: the probability of a word after “the capital of France is” differs from its probability across the whole corpus. This elementary difference is crucial. An expression’s global frequency does not by itself determine its probability in a question comparing two ideas.

For a sequence of symbols, the chain rule allows the joint probability to be factored into a product of conditional probabilities. This factorization is not a theory of meaning; it is a probabilistic identity. An autoregressive implementation attempts to estimate each factor using the available prefix. [R02, R03]

$$p(x_1, \dots, x_T) = \prod_{i=1}^T p(x_i \mid x_1, \dots, x_{i-1}).$$

A long sequence can have very low probability even if every step is banal. If each choice has probability one half, a sequence of one hundred choices has probability 2 to the power minus one hundred. The improbability of the entire sequence does not imply that it contains a discovery. This is why a document’s raw surprisal cannot be compared without controlling length and the granularity of representation.

The informational surprisal of an outcome is the negative logarithm of its probability. Choosing a base-two logarithm expresses information in bits. An outcome with probability one half has one bit of surprisal; one with probability one thousand in a million has approximately ten bits. These values describe rarity relative to the chosen dis-

tribution, not the outcome’s epistemic or moral importance. Shannon formulated communication theory precisely by separating the transmission problem from the semantic interpretation of messages. [R02]

$$s(x)=-\log_2 p(x).$$

A random identifier can have high surprisal without saying anything interesting about the world. A simple equation can have low textual surprisal and important consequences. Moreover, a false statement or typographical error can be improbable. A detector based only on surprisal would therefore also favor noise. Eliminating noise requires additional criteria; at that point, the detector is no longer measuring probability alone.

Perplexity is the exponential of the average logarithmic loss over symbols. It partially normalizes the effect of length and is useful for evaluating prediction. But comparisons between models with different tokenizations require caution: an average over tokens does not have the same effective unit when one model splits a word into two parts and another into five. In addition, perplexity depends on the context under which it was calculated. Saying that an idea has “low perplexity” without specifying the text, tokenization and conditioning is insufficient for a reproducible analysis.

The novelty problem is often classificatory rather than generative. We want the probability of a label or relation after seeing the contribution and prior work. This is a different distribution from the contribution’s probability in ordinary language. A text can be highly improbable yet correctly classified as incoherent. A highly predictable text can contain a small but detectable difference relative to explicit prior work.

$$D_{\theta}(x,K,C)=P_{\theta}(Y=1 \mid x,K,C).$$

$$G_{\theta}(x)=P_{\theta}(X=x \mid \text{generation prompt}).$$

The two expressions define different probabilistic objects that cannot be substituted for one another. Their values might coincide accidentally; that coincidence would not make the tasks identical. The generation distribution places weights on texts; the classification distribution

places weights on labels conditioned on the text and reference. The argument that a generator favors familiar continuations does not, without further steps, demonstrate that the same system cannot compare a contribution with a library.

Bayes also shows why a novelty signal can mislead when discoveries are rare. Suppose, in a constructed example, that one thousand out of a million candidates are genuinely new under the chosen criterion. The detector retains nine hundred of them but incorrectly classifies as new one percent of the nine hundred and ninety-nine thousand old candidates. We obtain nine hundred correct cases and nine thousand nine hundred and ninety false alarms. Only approximately 8.26% of the selections are genuine novelties.

$$\text{PPV} = \frac{\pi \text{TPR}}{\pi \text{TPR} + (1-\pi) \text{FPR}}.$$

Here π is the prevalence of novelty, TPR is the detection rate for new cases and FPR is the false-alarm rate among old cases. The formula follows directly from joint probabilities. It does not say that the detector is useless, but that performance must be judged relative to prevalence and verification capacity. In a discovery filter, the cost of a few false alarms may be acceptable if it prevents the loss of a rare contribution. Choosing the threshold is an operational decision, not an automatic consequence of the score.

2.2. What Learning Optimizes and What It Does Not Promise

Consider a true distribution P and an estimated distribution Q . If we evaluate predictions using logarithmic loss, the average cost of Q decomposes into the entropy of P and the Kullback–Leibler divergence between P and Q . The proof is algebraic: we add and subtract the logarithm of P in the sum defining the loss. The entropy term does not depend on Q ; the divergence term is nonnegative.

$$E_{X \sim P} [-\log Q(X)] = H(P) + D_{\text{KL}}(P \| Q).$$

In the ideal case where the model family contains the correct distribution and optimization finds it, the solution is $Q=P$. It does not place all the mass on the most frequent outcome. On the contrary, it also repro-

duces the probabilities of rare events. Confusing minimization of logarithmic loss with exclusively memorizing the distribution’s mode means confusing distribution estimation with a decision rule that chooses only the most probable outcome.

The practical situation is more difficult. Data are finite, model capacity is limited, optimization is imperfect and the test distribution may differ from the training distribution. A rare event may receive too few examples for the model to learn the relevant structure. In an objective averaged over all examples, improvement on frequent cases may numerically dominate improvement on a rare class. This observation is an argument about error allocation, not a theorem that rare cases are impossible to learn.

A model can learn a shared rule that also covers rare cases. If frequent examples enable it to learn addition, it can calculate a sum absent from the data. If it learns only local associations between expressions and labels, generalization will be weaker. The difference cannot be deduced solely from the objective. The same objective can admit solutions with different internal strategies. Identifying the strategy requires tests that prevent solving the task through superficial correlations.

The model distribution must also be separated from the decoding procedure. Choosing the most probable symbol at each step, sampling and search with multiple hypotheses can produce different behaviors from the same probabilities. Locally choosing the most probable symbol does not guarantee the globally most probable sequence. Increasing temperature can diversify outputs, but it does not turn the distribution into a novelty verifier. Textual diversity and validation of a difference are separate problems.

An evaluator answering “new” or “old” technically produces tokens. This observation does not imply that it cannot perform a comparison. An equivalence-checking program also ultimately produces bits, without the nature of its output exhausting the nature of its computation. The relevant question is what information and operations determine the output. Reducing a competence to the form of its final instruction does not constitute a sufficient functional analysis.

At the same time, a high probability for the token “new” must not be confused with a well-calibrated epistemic probability of historical priority. It may reflect prompt conventions, preferences introduced through post-training or the fact that hesitant answers were rewarded less. Calibration must be measured on clearly defined events. Research by Guo and colleagues shows, in a neural-classification context, that numerical confidence and empirical correctness can diverge and that post-processing can improve calibration. This does not imply a guarantee for LLM novelty judgments. [R18]

A rigorous protocol keeps the internal score, textual verdict and quality of justification separate. An answer can be correct with poorly calibrated confidence, incorrect with a fluent justification, or supported by a justification that identifies the correct difference but aggregates it into an unsuitable score. Reducing all these dimensions to a single percentage prevents understanding the failure mechanism.

2.3. Relevant Information and the Limits of Compression

Entropy measures a variable’s average uncertainty. Mutual information measures how much observing one variable reduces uncertainty about another. In our problem, X can represent the original document, Y the novelty label under a fixed criterion and Z the representation the system constructs. We are interested not only in how much Z retains about X . We want to know whether it retains what is needed for Y .

A representation can eliminate almost every detail of the document and still be sufficient for a narrow question. If we want to know whether a circuit ever outputs one, a correctly computed bit can summarize the answer. If we later want to know which inputs produce one, that bit is no longer sufficient. Good compression is not an absolute property. It depends on the family of future questions.

For a chain in which Z is derived only from X and the decision uses only Z , processing cannot create information about Y that was missing from the original observation. The intuition can be proved without metaphysics: any rule operating on Z can be simulated by a rule that receives X and first computes Z . Consequently, the best evaluator with

access to X cannot be worse than the best evaluator limited to Z , provided resources and decision classes impose no further restrictions.

This formulation leaves room for an apparently paradoxical fact. A particular model can perform better on a summary than on the full text. No additional information about the truth of the label has been created; its accessibility to a limited evaluator has changed. The summary can remove distractions, rearrange premises and reduce computational cost. The difference between a particular algorithm's performance and the optimal performance permitted by the information is fundamental.

The risk arises when the summary removes not noise but the relevant difference. Suppose two proposals have the same thematic description and differ only in a quantifier: "for every input" versus "for almost all inputs." If the representation retains the topic and generic result, the two become identical. This is no longer merely a matter of evaluator preference. The information needed for the distinction has disappeared from the channel it receives.

At the engineering level, this observation suggests that simply increasing vector dimensionality is not a complete solution. A large representation can ignore a relevant variable, while a small one can retain it explicitly. What matters is the representation objective, the invariant being pursued and the existence of a path back to the source. A novelty system should be able to say not only "these documents are similar," but also "in what respects we decided to consider them similar and what we removed in the process."

This is where information theory becomes a design discipline. It does not give us a universal metric of originality, but it forces us to ask what differences are preserved, what questions remain answerable and where additional access to the original documents is needed. Novelty cannot be deduced from the number of bits of surprisal. Conversely, losing certain semantically decisive bits can make its detection impossible for a given architecture.

Chapter 3. What a Language Model Learns and How It Comes to Judge

3.1. From Tokens to a Conditioned Procedure

A language model does not receive ideas as ideal objects already separated from their formulation. It receives a sequence of discrete units called tokens. Tokenization can split words, keep mathematical symbols separate and represent two similar formulations differently. Before any judgment of originality, there is therefore an encoding channel. This observation does not automatically imply loss of meaning, but it introduces a first control variable: task difficulty must not be confused with the difficulty of representing the text in the tokenization used.

Tokens are transformed into vectors. In the Transformer architecture, attention mechanisms allow information from different positions to be combined using weights dependent on the current representations. Successive layers modify these representations, and the output produces scores for continuations. The paper that introduced the Transformer describes this organization in the context of sequence modeling. The architecture does not demonstrate general reasoning or originality-assessment capabilities. [R03]

For our problem, attention must be understood functionally. It can allow a condition in one paragraph to influence interpretation of a guarantee in another. But a communication path between positions does not guarantee that the model tracks the correct condition. An architecture can express a computation without the learned weights implementing it reliably. Saying “the model can attend to the entire document” is a claim about structural access; saying “the model compares all relevant assumptions” is a claim about behavior.

A programmer can view an LLM as an approximate interpreter of a task formulated in text. The prompt specifies an incomplete contract, examples and context supply data, and the model produces an output

according to learned regularities. This analogy is useful provided it is not taken literally. There is not necessarily a stable internal separation between parser, planner, memory and verifier. Nevertheless, in external evaluation we can construct separate tests for understanding the instruction, extracting claims, retrieving prior work and applying the criterion.

In-context learning is adaptation of behavior using examples in the prompt, without updating parameters during that interaction. Brown and colleagues documented important few-shot results for a large autoregressive model. These results show that performance is not fully described by the question “did this exact example exist in training?” They do not, however, guarantee learning any new rule from a few examples. [R04]

In a novelty test, the context can define the reference world itself. We can supply a library of three mechanisms and ask the model to determine whether a fourth is equivalent to one of them. If the mechanism names are generated randomly after the model is fixed, recognition by reputation becomes harder. The model must use local definitions. Success would demonstrate comparison competence in a controlled world; failure would indicate a concrete obstacle without immediately identifying which internal layer produces it.

The term “semantic representation” must be used modestly. A vector can correlate strongly with useful meanings or relations without constituting a complete, stable and interpretable description of the idea. In a retrieval system, we care whether vector proximity brings relevant documents. In a verifier, we care whether differences between assumptions and conclusions are preserved. The same representation can be suitable for the first operation and insufficient for the second.

It would be a design error to treat the thematic vector as the identity of the contribution. Two papers about the same problem must be close for retrieval to work. Precisely for that reason, their proximity should not be used directly as evidence that no difference exists. A representation optimized to bring relevant documents together may compress distinctions that the final comparison must reintroduce. A good semantic neighbor is not automatically prior work that anticipates the contribution.

Consequently, evaluating LLMs must begin with a functional diagram of the available information, not assumptions about what the model “knows.” What text does it receive? What parts are truncated? What instruction defines novelty? What sources are added? What output form is requested? An incorrect answer can originate at any of these points. Without this decomposition, the explanation “the model repeats the past” risks becoming a universal label applied to very different failures.

3.2. Generalization, Memorization and Priorities Introduced through Post-Training

Memorization and generalization are not mutually exclusive states of a model. A system can memorize certain sequences while also learning regularities applicable to new sequences. For novelty detection, both capacities can be useful or problematic. Memorizing prior work helps reject a rediscovery. Memorizing a paper’s reputation can contaminate a retrospective test. Generalizing a comparison rule helps identify differences; overgeneralizing a category can erase the very exception that matters.

Showing that the model can produce familiar fragments is not enough to conclude that every verdict retrieves fragments. But absence of literal reproduction does not prove conceptual comparison either. A model can reformulate a memorized association. Evaluation must construct situations in which memorization and comparison make different predictions: changed names, reversed relations, assumptions moved between candidates and prior work, and cases in which familiar wording conceals the wrong conclusion.

A useful test is a controlled reversal of the world. In one condition, the library contains method A and the candidate proposes B. In another, the same pair is reversed and the library contains B. If the label depends on the reference, the verdict must change coherently. An evaluator that always declares B newer because B uses sophisticated vocabulary fails to respect the task’s local definition. This manipulation is more informative than simply requesting further explanations.

Models used as assistants are generally not fully described by autoregressive pretraining. There may be stages of supervised fine-tuning and preference optimization. Ouyang and colleagues describe a procedure combining human demonstrations, answer comparisons and learning from feedback. Rafailov and colleagues propose a direct preference optimization method without constructing the entire reward-learning loop in the same way. These papers show that the final policy can be modified to favor particular types of answer. They do not establish that the preferences introduced are necessarily hostile to novelty. [R05, R06]

In our problem, post-training is a causal candidate that must be separated from pretraining. If human evaluators prefer cautious, explicit and conventional formulations, the system may learn to produce or appreciate them. If evaluation examples reward mentioning standard terms, those terms may become cues. But these are hypotheses about data and objectives. They cannot be demonstrated merely by observing that an answer is conservative.

We could compare a base model with its fine-tuned version, but the comparison does not automatically isolate a single mechanism. Instruction-following ability, style, answer length and context use may all change simultaneously. A causal evaluation requires either more controlled interventions or a limited interpretation: “a difference between two training procedures,” not “the pure effect of a preference for conformity.”

Model-generated explanations are useful as objects to verify, not as direct access to the verdict’s internal cause. A model can produce a plausible explanation after selecting its score from other cues. To test whether prior work was decisive, we can remove or modify it and observe the behavioral change. To test whether a difference was understood, we can request a prediction that depends on it. Intervening on information is stronger than mere textual introspection.

One verification strategy separates verdict from witness. The model must identify an earlier passage anticipating the contribution or a case in which the candidate and prior work behave differently. The witness can be checked independently. If the score says “new” but the witness is false, we have a detectable error. If the witness is correct

but the score too low, the problem shifts to the relevance or aggregation criterion. This separation reduces dependence on explanatory fluency.

Finally, a model's competence must not automatically be attributed to the entire system, or vice versa. An LLM with access to search, computation and verifiers can perform differently from the same LLM in isolation. A contribution found by a symbolic verifier does not prove that the model solved the problem alone. But using a verifier does not invalidate the system's usefulness either. Reporting must specify who proposed the comparison, who executed it and who certified the result.

3.3. Evaluator Failures That Resemble One Another Only on the Surface

An evaluator can miss an idea because prior work is absent. This is a coverage failure. It can receive the prior work but omit the relevant paragraph. This is a failure of effective context access. It can read the paragraph and confuse the quantifiers. This is an interpretation failure. It can identify the difference and mistakenly judge it unimportant. This is a failure to apply the criterion. It can produce all the correct elements but an inconsistent final score. This is an aggregation or instruction failure.

The distinctions matter because the remedies differ. Expanding the search engine can improve coverage but does not resolve confusion between “there exists” and “for all.” A more explicit prompt can improve adherence to the criterion but does not supply the missing document. A larger model can reduce some errors but cannot recover information permanently removed by an intermediate summary. Without diagnosis, increasing inference cost risks producing only longer explanations of the same error.

There are also failures of information positioning. Lost in the Middle reported performance dependent on the position of relevant information in long contexts for the models and tasks tested. This does not demonstrate that all future models will have the same problem. It does, however, provide a concrete reason to control document order and the location of decisive prior work in experiments. [R16]

A novelty filter is especially exposed because the important difference can be small relative to the text's total length. If the document contains ten pages of familiar context and one sentence changing the scope of the guarantee, success depends on the operational attention given to that sentence. Correctly summarizing most of the document is not enough. A local error can reverse the global verdict.

Let us construct a simple case. The prior work sorts a collection in memory, while the candidate claims the same order using sequential access to external data. A thematic evaluator may observe that both are sorting algorithms. An operational evaluator must compare the access model and resources. If the task asks about novelty in the computed function, the two may be equivalent. If it asks about novelty in the execution model, the difference may be central. The verdict cannot be judged without knowing the evaluation contract.

Another constructed case concerns negating an assumption. One text says that the procedure requires an authenticated channel. Another claims that this requirement can be replaced by a different assumption. A model normalizing the second formulation to the first can make the text more familiar but less faithful. This "correction" is not necessarily intentional. It may result from the most probable interpretation. Nevertheless, for detecting unusual contributions, intention is secondary; preserving the difference is what matters.

The tendency to produce evaluations matching the user's tone must also be controlled. If the prompt asks "why is this idea revolutionary?", the task already implies a verdict. If it asks "why is this merely a banal variation?", it creates another pressure. A benchmark comparing only affirmative versions of the same question does not measure epistemic stability. Favorable, neutral and skeptical formulations must be tested on the same content without suggesting that any one represents the truth.

This chapter's conclusion is functional. "The LLM judges novelty" is not an elementary operation. It is a chain of transformations and choices, some implemented by the model, others by the surrounding application. The familiarity hypothesis becomes scientifically useful only when located within such a chain and when an intervention can distinguish the proposed mechanism from alternative explanations.

Chapter 4. An Operational Semantics of Contributions

4.1. The Idea as a Structure of Claims, Not a Topic

A topic is a territory of discourse. A contribution is a definite change within that territory or a change in its rules. “Artificial intelligence for research” names a topic. “A verifier that detects contradictions between the stated assumptions of two experiments” describes a function. “A verifier that does this without access to raw data, under conditions X” introduces a possible difference. For novelty evaluation, moving from topic to structure is essential.

An operational representation can describe the problem, assumptions, mechanism, guarantees, costs and evidence. Not every field must be completed in every document. A missing field is itself information: if a proposal does not state its assumptions, the evaluator should not invent them to make it comparable. It can formulate multiple interpretations and mark which verdict depends on each.

We will schematically write a contribution as a structured object. The notation is an analytical tool, not a requirement that every idea fit one universal form.

$$x=(P,A,M,G,R,E).$$

P denotes the problem, A the assumptions, M the mechanism, G the claimed guarantees or results, R the resources and E the evidence supplied. The difference between two contributions can occur in any component or in the relation between components. The same guarantee under weaker assumptions is not the same claim. The same mechanism with a different proof can constitute another type of contribution. The same results obtained with fewer resources may have operational novelty even if the final function remains identical.

To avoid turning the scheme into a rigid ontology, every field must be linked to original passages. The representation must not replace the document without the possibility of auditing it. An evaluator can say

“I extracted assumption A from this passage” and retain an alternative version if the text is ambiguous. Structuring aims to preserve differences, not force them into familiar vocabulary.

Negation, modality and quantification are especially important. “May improve” is not equivalent to “improves in every case.” “There exists a parameter set” is not equivalent to “for every parameter set.” “Under the independence assumption” is not a stylistic note. These elements can be short in the text and large in their consequences. A summary that retains the nouns but loses logical operators can remain thematically correct and semantically useless for novelty.

A contribution structure can be viewed as a small graph. Nodes represent claims, assumptions, results and sources. Edges can indicate dependence, justification, contradiction or the transformation of an assumption. The graph must not be confused with a proof: linking two nodes with a “justifies” edge does not certify the relation’s validity. It does, however, provide precise places where a verifier or expert can intervene.

This representation enables better questions than “is the idea new?” What guarantee is absent from the prior work? What assumption has been removed? What observation would distinguish the mechanisms? What claim is merely a reformulation? What part of the contribution depends on an unverified assumption? Each answer can be evaluated separately. We do not automatically obtain truth, but we obtain a better surface for correction.

4.2. Equivalence, Anticipation and Residual Difference

Equivalence is a stronger relation than resemblance. Two objects can resemble one another without being equivalent; they can be equivalent under one criterion and different under another. For functions on a finite domain, extensional equivalence means identical outputs for every input in that domain. For scientific papers, the criterion can be much more complex. A small distance in vector space therefore cannot be treated directly as proof of conceptual equivalence.

We will use $x \text{ equivalent}_C y$ to denote that two contributions are equivalent under criterion C. If the relation is genuinely an equivalence relation, it must be reflexive, symmetric and transitive. Many similarity scores do not satisfy these requirements when thresholded. We can have A similar to B and B similar to C without A being similar to C. Constructing classes of ideas through the transitive closure of such a relation can group together contributions that should not have been confused.

Anticipation is often asymmetric. Prior work can already contain all the relevant elements of the candidate while also including additional results. In this case, the candidate need not be identical to the prior work for its novelty claim to be weakened. We can represent the relation through inclusion of relevant content, provided we specify what content means and which inference rules are permitted.

$$\text{novel}_C(x, K) = 1 \Leftrightarrow \neg \exists k \in K : \text{Ant}_C(k, x).$$

Here, $\text{Ant}_C(k, x)$ means that prior work k anticipates contribution x under criterion C. This is a working definition in a closed world, not a legal definition of novelty or a guarantee concerning the entire history of ideas. In real applications, several sources may jointly contribute to what appears new in a candidate. If we allow any combination of prior work, we must establish when the combination is considered already available and when constructing it is itself the contribution. Without such a rule, the evaluator can declare every idea “a combination of known things,” which no longer discriminates anything.

A useful concept is residual difference. After identifying the closest prior work, we extract what remains unanticipated under the stated criterion. The difference can be empty, clear or uncertain. It may consist of a removed condition, an introduced type of object or a new relation between old components. “Residual” does not mean minor; it means what remains after an explicit comparison.

A constructed example: prior work A solves problem P for acyclic inputs. Prior work B handles cycles, but only in a restricted case. Candidate C claims a solution for all inputs in a larger class. A summary saying “combines A and B” is insufficient to deny novelty. We must show either that applying the prior methods together already pro-

duces C's guarantee or that C does not actually deliver its claimed guarantee. In the first case, the novelty claim is anticipated; in the second, validity is challenged. These are different arguments.

An instructive difficulty arises in formal mathematics. If we define knowledge as all the logical consequences of accepted axioms, every provable theorem was already implicitly contained in the axioms. No new proof would then expand the set of consequences. This definition may be logically useful, but it does not describe the epistemic progress of a resource-limited agent well. Having an implicit consequence is not the same as having identified or proved it within a practical time.

We can introduce a resource-dependent closure: the consequences that specified procedures can obtain from K within budget B. A new proof can expand what is accessible even if it does not expand ideal logical consequence. This idea must be used cautiously because the result depends on the procedure and representation. Nevertheless, it reveals a type of novelty that simple semantic inclusion misses: turning an abstract possibility into effectively usable knowledge.

In computer science, the same phenomenon occurs when a compilation method turns a correct but impractical computation into an efficient one. The computed function can remain identical while the contribution is substantial. A detector identifying only new functions would miss it by definition, not through lack of intelligence. The criterion must therefore be chosen according to the type of progress we want to observe.

4.3. When the Idea Changes the Very Vocabulary of Comparison

The most difficult situation is not always a new value in a known field. Sometimes the contribution challenges the fields. Instead of proposing a better mechanism for the accepted problem, it redefines the problem. Instead of optimizing a guarantee, it suggests that the guarantee is unsuitable. Instead of adding a category, it changes the criterion by which categories were constructed. An evaluator requiring all ideas to conform to the same scheme can miss precisely this type of change.

It does not follow that every rejection of existing vocabulary is profound. Some reformulations produce no additional predictions, operations or clarifications. Others conceal incoherence. The real difficulty is that the novelty test must allow the criterion to be challenged without abandoning verification. The solution proposed here is evaluation at two levels: comparison within the current framework and reasoned examination of a change in framework.

At the first level, the system can say that the proposal introduces no new function under the existing criterion. At the second, it can ask whether the proposal exposes a limitation of the criterion, offers examples the framework cannot distinguish and introduces an alternative with verifiable consequences. A change of framework becomes more than a metaphor when it shows what can be observed or done differently. Conceptual originality is thus neither reduced to lexical distance nor exempted from the obligation to produce intelligible differences.

A constructed example for programmers is a change in the cost model. Two algorithms may be equivalent if we count only arithmetic operations but different if data transfer is the dominant cost. Proposing another cost model does not change the computed function; it can, however, change the optimization question and the ordering of preferred solutions. The evaluator must distinguish a merely convenient change of metric from a change justified by application conditions.

Another example concerns privacy. If the system describes two procedures only through accuracy and time, it cannot evaluate their difference in information disclosure. Introducing an explicit adversary model can show that the old representation was insufficient. We need not claim that every new adversarial model is historically original. The example shows only that novelty can consist in adding a relevant dimension, not a new value on old dimensions.

In an automated system, an ontology change must be managed without retroactively rewriting the data. One solution is to preserve multiple interpretations and a translation relation between them. The contribution remains linked to its original formulation, and the evaluator can compare results under alternative criteria. If all interpretations are compressed into one at the outset, the possibility of revision

disappears. If they are retained without limit, the cost becomes uncontrollable. The problem is how to allocate the budget among plausible interpretations.

An important research direction appears here: detecting representational insufficiency before detecting novelty. A system might observe that two texts are very close in the current representation yet contain incompatible claims or explicit differences the scheme cannot encode. Instead of declaring them identical, it could request an expanded scheme or human verification. This is an architectural proposal, not a capability demonstrated in this book.

Such a mechanism must also be tested against misuse. If every lexical difference triggers a new ontology, the system becomes useless. The escalation criterion should depend on a concrete question to which the interpretations give different answers. In this way, representational change remains connected to consequences rather than a vague impression of depth.

This chapter closes with a warning and an opportunity. We cannot detect every type of novelty in a space that has removed its constitutive differences from the outset. But we can build systems that declare their criteria, preserve sources and allow the representation to be revised. Such discipline does not promise infallible recognition of conceptual revolutions. It at least prevents confusing the limits of a scheme with the limits of what can be new.

Chapter 5. How Familiarity Can Become a Substitute for Novelty

5.1. Convenient Cues and Evaluation Shortcuts

Suppose that in an evaluator's data, contributions regarded as original are more often presented with a particular structure: an important problem, a limitation of previous work, a new mechanism and convincing results. The structure may be good communication practice. But it can also become a statistical cue for the label "original." A model learning the cue can often succeed without actually comparing the mechanism with prior work. When the cue and the truth diverge, performance may deteriorate.

This is a hypothetical shortcut-learning mechanism. It does not imply that rhetorical structure is useless or that the model uses no content at all. Most likely, combinations of cues must be investigated in a concrete application. A score can depend partly on the real difference and partly on fluency, vocabulary, length, authority or fit with a familiar domain. The experimental question is how much each factor contributes and under what conditions one substitutes for another.

The rhetoric of novelty can be separated from the stated novelty. In a controlled pair, the same content can be presented as "we introduce a revolutionary framework" or "the procedure differs from prior work in the following condition." If the score changes without the claims changing, there is sensitivity to packaging. This sensitivity should not immediately be called an effect of training frequency. It may come from instructions, post-training, implicit clarity criteria or stylistic preferences. Identifying the cause requires further interventions.

Another cue is terminological density. A proposal combining fashionable terms can seem more advanced than a plain explanation of the same operation. But removing all technical terms is not a good solution: some are necessary for precision. The experimental manipulation must preserve information, not merely the general impression.

Two versions declared “equivalent” by the very LLM being evaluated may contain subtle differences. The manipulation’s equivalence must be independently checked or constructed in a formal domain.

Familiarity can also work in reverse. An unusual text may receive a high score precisely because it seems different. A superficial evaluator may confuse strangeness with originality, not only originality with familiarity. The hypothesis must be formulated to allow this possibility. There is no single mandatory direction of bias. A model may prefer familiar clarity in one context and extravagance in another, depending on what it has learned is meant by “creative.”

Source prestige is another distinct variable. If the author’s or institution’s name changes the verdict for the same text, we have an attribution effect. This is not identical to the effect of an idea’s frequency. An author can be familiar without the mechanism being old. A model may use the name as a prior about probable quality, and this prior can be reasonable for some tasks and unsuitable for anonymous novelty evaluation. The evaluation contract must state whether such information is admissible.

The problem of convenient cues is not ontologically specific to artificial intelligence. Any limited evaluator can use heuristics. Automation’s practical difference is that the same heuristic can be applied at scale with a uniform style of justification. This statement is a deduction about multiplying a decision rule, not an empirical estimate of human communities’ behavior. Comparing humans and models requires the same texts, the same sources and the same verification resources.

An important consequence is that robustness to reformulation should not be measured only on comfortable paraphrases. If all variants are produced by the same model in the same style, the test may confirm invariance over a very narrow space. Technical, neutral, minimal and explanatory variants are needed, along with controlled changes in order and terminology. At the same time, we must not turn resilience to defective prose into the definition of novelty. A detector may legitimately need clarification.

5.2. Semantic Smoothing: When Interpretation Repairs the Very Exception

We will use the term semantic smoothing for a family of transformations that reduce variation between texts. Some remove irrelevant variations, such as the order of equivalent explanations. Others normalize terminology. Still others choose the most probable interpretation of an ambiguous passage. Smoothing is not a defect by definition. Without certain invariances, the system would mistake every phrase for a new contribution.

The risk arises when the system smooths a relevant variation. A text claims that the mechanism works without a usual condition. Because this claim is unusual, the model may interpret it as an omission or imprecise formulation and reintroduce the condition in the summary. The result becomes more plausible but less faithful. The very claim to be evaluated has been transformed into a conventional version.

We can describe this phenomenon as a competition between fidelity and the plausibility prior. When interpreting noisy text, choosing the probable meaning is often useful. When evaluating an idea claiming to overcome a limit, the same choice can be dangerous. The system must distinguish between “the author probably meant something ordinary” and “the author explicitly claims something unusual that must be checked.” The second preserves the object of criticism; the first may eliminate it.

A constructed example uses a quantifier. The prior work guarantees that for every input there is a suitable parameter. The candidate guarantees the existence of a single parameter that works for all inputs. The two propositions may contain the same nouns and almost the same words. The order of quantifiers, however, radically changes the claim. A summary such as “there are parameters that ensure operation” can make them indistinguishable. This is not ornamental loss; it is logical loss.

A second example concerns causality. Prior work reports an association between two variables. The candidate claims that intervening on one changes the other under certain conditions. A summary saying “shows a relationship between variables” preserves the topic and re-

moves the difference between association and intervention. The candidate may be wrong in its causal claim. But if the difference is erased, the system no longer even knows what should be validated.

A third example concerns the domain. Two methods have the same formula, but one is defined on finite objects while the other claims an extension to an infinite space. If the representation retains the formula and ignores the domain, it will overestimate equivalence. Again, we need not assume that the extension is correct to establish that the claims differ. Detecting the claimed contribution and validating it must remain separate.

Smoothing can occur in claim extraction, summarization, translation, embeddings, keyword selection and search-query reformulation. In a multistage chain, a small difference can be lost repeatedly. A stronger final model cannot reliably recover it if all the representations it receives are already identical. Evaluating only the final answer can therefore conceal the error's origin.

An appropriate test compares three paths. The first preserves the original text. The second supplies only the compressed representation. The third supplies the representation and access to the decisive original passages. If performance falls in the second and recovers in the third, there is evidence that loss or inaccessibility of intermediate information matters. To attribute the decline specifically to a semantic collision, we must verify that the representation no longer encodes the difference, not merely that the evaluator finds it harder to use.

An auxiliary task is also useful: after compression, can a verifier recover the answer to the question that distinguishes the candidate from prior work? For the quantifier example, the question is whether the parameter may depend on the input. If the summary does not permit an answer, we have a direct measure of relevant information loss. A generic summary-fidelity score can miss this defect because most of the content was correctly preserved.

Not every detected loss justifies abandoning compression. The cost of retaining everything must be compared with the value of the difference and the frequency of the cases. An architecture can retain summaries for navigation and sources for verification, include special fields for assumptions and exceptions, or escalate cases in which two

interpretations produce different consequences. The goal is not the absence of all abstraction, but explicit control of abstractions that affect the verdict.

5.3. Attraction to Prior Work and Confirmation Loops

Retrieving related documents seems the natural solution to the novelty problem. But a system may search through a reformulation that has already removed the candidate's difference. It will then retrieve precisely the familiar literature and confirm its own initial interpretation. Not because the literature demonstrates identity, but because the query excluded what needed investigation. This is a possible architectural confirmation loop, not an inevitable property of every search engine.

Suppose the proposal concerns a guarantee without global synchronization. The generated query becomes “guarantee in synchronized systems” because that is the familiar category. Retrieved results contain abundant thematically related prior work. The evaluator concludes that the idea is well known. The final error looks like a novelty judgment, but its cause arose in translating the search request. A provenance log must preserve queries as well as the final documents.

There is also a symmetric risk. If the query uses only the candidate's unusual vocabulary, the engine may miss prior work described in other terminology. The system declares novelty because it found no lexical matches. The solution is not to choose literal or semantic search exclusively, but to maintain several paths: exact terms, synonyms, mechanism, assumptions, relations and consequences. Each path must be evaluated for its contribution to coverage.

Consensus among several models does not automatically solve the problem. If all receive the same summary or use the same index, their errors may be correlated. Diversity of model names does not guarantee diversity of information or criteria. An ensemble can reduce random variation while reinforcing a shared bias. Demonstrating a benefit requires measuring error complementarity, not just final agreement.

An adversarial role can be useful: one evaluator searches for prior work while another tries to construct a case distinguishing the candidate. Nevertheless, the dispute must not be treated as a magic method for producing truth. Two agents can generate persuasive but false arguments. Value arises when arguments contain verifiable objects: sources, locatable quotations, calculations, counterexamples or explicitly separated assumptions. Debate without witnesses risks being merely rhetorical competition.

The institutional loop appears when verdicts are reused as data. If accepted ideas become positive examples without auditing, the system may learn its own preferences. If rejected ideas are not retained, there is no longer a way to estimate what was lost. In such a process, the absence of observed errors may reflect a lack of observation rather than correct selection. Archiving rejections and randomly reevaluating them are therefore epistemic components, not merely administrative features.

It should not be inferred that every use of synthetic data inevitably produces collapse. Studies of recursive training show important dependencies on whether data are replaced or accumulated. These results will be discussed separately. Here the argument is narrower: an evaluation that becomes its own ground truth can lose its ability to be corrected. This possibility exists even without retraining if earlier decisions are treated as authoritative precedents in every new round.

The familiarity hypothesis must therefore extend from the model to the information circuit. Sometimes the problem is a learned prior. At other times it is a search query, a compressed representation, an aggregation rule or a data-retention policy. A responsible system does not attribute every error to “model hallucination,” but preserves enough traces to identify the transformation that made the difference invisible.

Chapter 6. Genuine Impossibilities, Conditions of Possibility and Proofs

6.1. Missing Information and the Impossibility of Absolute Priority

Let us rigorously define an evaluator as a procedure that receives observations O and produces a verdict. Observations may include the candidate document, accessible sources, search-engine responses and tool outputs. The complete world may also contain inaccessible documents. If two worlds produce exactly the same observations for the evaluator, the procedure cannot distinguish them on the basis of those observations. This is an elementary identifiability limit.

We construct two worlds. In world W_0 , contribution x has no prior work before time t . In world W_1 , prior work exists in a document the evaluator cannot access. All accessible documents are identical in the two worlds. The training history and evaluator state are also identical. The correct historical verdict differs, but the procedure's input does not. A deterministic evaluator will produce the same answer in both worlds and be wrong in at least one.

Randomization does not remove the problem. A randomized evaluator will produce the same answer distribution in the two worlds because its observations have the same distribution. It cannot have probability one of being correct in both. If we assign equal probabilities to the two worlds, the best verdict without further information cannot exceed probability one half of identifying the correct world. With other priors, it can favor the more probable world, but it does not obtain informational certainty.

This proof does not say that searching for prior work is useless. It says that an absolute guarantee requires assumptions about completeness of access. In a finite, fully enumerated library, we can check the absence of an element under a decidable criterion. In an open history of

knowledge, we can produce a conclusion conditional on the sources examined. The difference between these situations must be expressed in the final report, not hidden in the fine print.

The limit applies to humans, search engines and hybrid systems. It is not a special property of language models. It therefore cannot be used to demonstrate that LLMs are weaker than humans. A relative comparison must supply the same sources and budget or measure the access advantage separately. Otherwise, competence and information are confused.

A more modest guarantee can be formulated: “we found no prior work in K using procedure P, and the procedure’s coverage was evaluated on a family of similar cases.” This is a verifiable claim. It is not equivalent to historical absence, but it may suffice for a practical decision. Confidence depends on the domain, indexing and the procedure’s ability to find prior work with different vocabulary.

A negative search result has value only relative to search sensitivity. If the procedure frequently misses known prior work, finding nothing for the candidate is weakly informative. A useful test involves concealing the identities of papers in a controlled way and checking whether they can be retrieved from their contributions. This test does not prove historical completeness, but estimates a component needed to interpret the absence of results.

The formal conclusion is narrow and strong: no evaluator can universally certify the absence of inaccessible prior work without additional assumptions. This proposition does not imply that recognizing a new difference relative to an accessible library is impossible. An unjustified move from the first to the second would turn an informational limit into a denial of comparison.

6.2. The Didactic Theorem of Sufficient Representation

Consider a set X of objects, a deterministic labeling $Y: X \rightarrow \{0,1\}$ and a representation $\varphi: X \rightarrow Z$. The question is whether there exists a classifier $g: Z \rightarrow \{0,1\}$ that perfectly decides the label using only the representa-

tion. The following result is elementary and is not presented as a new theorem of this book. It makes the mechanism of difference loss explicit.

Proposition. A perfect classifier g with $Y=g\circ\varphi$ exists if and only if Y is constant on every fiber of φ . A fiber is the set of all objects receiving the same representation.

$$\varphi(x)=\varphi(x') \Rightarrow Y(x)=Y(x').$$

The proof of necessity is immediate. If two objects have the same representation, g receives the same input and produces the same output. To be correct on both, their labels must be equal. The proof of sufficiency constructs g : for each representation z , we choose the common label of the objects in its fiber. The assumption that the label is constant guarantees that the definition does not depend on the chosen object. On representations not reached by φ , g can be defined arbitrarily.

The proposition is informational: the existence of the function g does not guarantee that it is computable or efficient on an infinite domain. In the finite case, we can construct a decision table. In the general case, computability remains separate from the information-preservation condition.

The consequence for novelty is precise. If a new contribution and an old one, under the same criterion and reference, map to the same representation, no evaluator restricted exclusively to that representation can be perfect on both. It does not matter how fluent it is, how many parameters it has or how long it reflects. It may guess better using prevalences, but it cannot recover the lost distinction from an identical signal.

There is, however, an often omitted condition: mere noninjectivity of the representation does not demonstrate failure. A representation can group many paraphrased texts, all with the same label, without losing anything relevant to the task. If φ removes only irrelevant variations, classification can remain perfect. The problem is not that two different texts have the same vector. It is that two texts with different labels provide the same information for the decision.

Moreover, the criterion and library must be included in the total input. The same contribution may be new relative to K1 and old relative to K2. A candidate representation, however good, is insufficient if the evaluator does not receive the reference. We can define the total object as $u=(x,K,C)$. The theorem applies to the representation of u . This clarification prevents a common error: treating novelty as a fixed label of a text, independent of context.

For a finite example, suppose we have four equally probable objects. The first two are old, the next two new. Representation Z has two values: one groups the first old object with the first new object, while the other groups the remaining two. In each fiber, the labels are equally mixed. Any classifier based only on Z is wrong on half the objects. If the representation changes so that each fiber contains only objects with the same label, the minimum error becomes zero, even though the representation remains compressed.

In the discrete probabilistic case, the minimum Bayes risk for binary classification from Z is the sum, over representations, of the minority class mass in each fiber. The formula simultaneously includes representational loss and any label noise.

$$R_Z^{\text{opt}} = \sum_z \min \{P(Y=0, Z=z), P(Y=1, Z=z)\}.$$

For equal class priors, the same bound can be expressed through the total variation distance between the class-conditional representation distributions. Total variation measures how different the distributions are; if they are identical, there is no discriminative signal.

$$R_Z^{\text{opt}} = \frac{1 - \text{TV}(P_{Z|Y=0}, P_{Z|Y=1})}{2}.$$

These formulas do not allow automatic estimation of an LLM's error from an arbitrary vector. The relevant distributions must be known or estimated, and internal representations are generally complex. They do, however, offer a conceptual target for experiments: we seek transformations that reduce label separability, not merely text length. We can measure this reduction directly in controlled finite worlds.

A corollary concerns the representation budget. With B bits, we can distinguish at most 2^B states. If the family of future questions requires distinguishing more relevant classes, some must be

merged. But a single binary question may require much less information than full reconstruction of the document. There is therefore no simple relation of the form “short summary, hence novelty impossible.” The limit depends on how many distinctions must be preserved for the permitted tasks.

Finally, source access changes the problem. If the evaluator receives Z and can request passages from X , it is no longer limited to Z . The theorem does not prohibit recovering the difference through a new observation. This observation underpins the architecture proposed later: compression for navigation can be combined with verifiable access to the original for decisions that depend on details.

6.3. Computability, Generalization and Constructive Counterexamples

Some novelty limits arise from computation rather than missing documents. If ideas are arbitrary programs and the criterion requires behavioral equivalence for all inputs, we enter general problems of program semantics. Rice-type results concern the impossibility of total algorithmic decision for nontrivial semantic properties of partial computable functions. The paper by Hemaspaandra and Rothe explicitly discusses the difference between such results and complexity analogues for restricted classes. [R25]

Delimitation is essential. Undecidable does not mean that no instance can be solved. It means there is no algorithm that halts and correctly decides every instance in the general class. Many concrete programs can be compared, many properties can be proved and restricted classes may be decidable. Invoking computability to declare every novelty detector useless would be an invalid extension of the result.

We can see the difficulty through a simple construction. From an arbitrary program M and an input w , we construct program P , which simulates $M(w)$ and returns zero if the simulation terminates. If it does not terminate, P produces no result. A universal evaluator able to decide whether P is equivalent to the constant-zero function on all inputs would decide whether $M(w)$ halts. Thus, a general equivalence

question can conceal the halting problem. The construction explains the limit without claiming that all novelty tasks have the same difficulty.

For Boolean circuits with a finite number of inputs, the situation differs. We can evaluate all 2^n inputs and compare truth tables. The procedure always halts. The cost may be high, but the problem is decidable. If the library is finite, we can check whether the candidate's function coincides with any stored function. This is a constructive counterexample to the claim that relative novelty detection is impossible in principle.

The system must also be distinguished from its linguistic component. An LLM can translate a description into a circuit, and an exact verifier can decide equivalence. Success of the whole system shows that LLMs can participate usefully in novelty detection, not that their translation is infallible or that the model alone proved everything. We can evaluate translation fidelity and verifier correctness separately. This separation makes the experiment more informative than a monolithic verdict.

There is also a simple counterargument to the thesis that a model can recognize only objects it has encountered. A comparison rule learned on old pairs can be applied to pairs generated later. If a model correctly reads truth tables and identifies a different row, that exact pair need not have existed in training. We are not claiming that every LLM does this robustly; we are saying that learning's historical origin is not a proof of impossibility.

No-free-lunch results are often invoked too broadly. In a world where unknown labels are arbitrary and independent of previous observations, no rule can systematically generalize better than another on average over all labelings. But real problems are not uniformly distributed over all possible functions. They have structure, regularities and criteria. The correct conclusion is that generalization requires inductive assumptions, not that generalization is impossible.

For novelty, the desired inductive assumption is not "the future always resembles the past." It may be "the comparison rule remains valid for new objects." A circuit comparator does not assume that the new circuit is identical to an old one; it assumes that gate semantics

and the input domain are stable. This difference between regularity of procedure and regularity of outcome is central. A system can use known rules to identify an unknown difference.

A clear map of limits emerges. Without information about inaccessible prior work, absolute priority is not identifiable. After a relevant distinction is lost, perfect classification from the remaining representation is impossible. For general programs, certain semantic comparisons are undecidable. Conversely, in finite worlds with explicit criteria, relative novelty can be decided exactly. The empirical question about LLMs begins between these limits: how well do they translate, compare, search and preserve information in tasks where a correct verdict is possible?

Chapter 7. What the Literature Shows and What It Cannot Demonstrate

7.1. Familiarity, Self-Evaluation and Perceived Creativity

The study Self-Preference Bias in LLM-as-a-Judge by Wataoka, Takahashi and Ri investigates evaluators' preference for answers familiar from the model's perspective. The authors report an association between lower perplexity and more favorable evaluations, more pronounced than in human judgments, including beyond the text's merely originating from the same model. The result supports investigating familiarity as an evaluation factor. It does not directly demonstrate rejection of radically new scientific ideas or by itself identify pretraining frequency as the sole cause. [R07]

This interpretive limit matters. Preference for an easily anticipated answer may reflect clarity or correctness in some tasks. To demonstrate a bias against novelty, we need cases in which familiarity changes while quality and conceptual difference remain controlled. A result about an evaluator's overall preference is one piece of the argument, not the entire argument.

In their paper on MT-Bench and Chatbot Arena, Zheng and colleagues examine LLM evaluation and discuss position, verbosity and self-favoring effects alongside reasoning limitations. The study shows both the practical usefulness of evaluators in the tested configurations and the need to control certain biases. High agreement with human judges on a conversational benchmark cannot automatically be transferred to historical priority or proof evaluation. [R08]

An experimental design can use these results as a starting point without exaggerating them. If answer order changes preference in a comparison, the order of candidates and prior work must be randomized. If length affects the score, it must be controlled separately from

information completeness. If the model favors its own style, comparisons between human- and model-generated ideas must separate content from the source or form of writing.

Si, Yang and Hashimoto conducted a study involving over a hundred NLP researchers in which LLM-assisted ideas were rated as newer than human ideas in the blind human evaluation used, with a result reported as statistically significant. The authors also discuss limitations involving feasibility, diversity and self-evaluation. The result contradicts a simplistic version of the claim that models can produce only texts perceived as banal. It does not establish the historical priority of all the ideas and is not a direct test of novelty-detection ability. [R09]

A major methodological distinction lies here: an idea can be judged original in a blind reading and still have obscure prior work. This is not a contradiction between the results of the two procedures. They measure different operations. Evaluating the impression of novelty may be relevant to communication or initial selection; checking prior work is necessary for stronger claims.

Gupta and Pruthi examined fifty AI-generated research documents with thirteen experts. In their sample, 24% were identified as paraphrases of or substantial borrowing from existing papers, with additional checks of the identified links. The study illustrates the difference between an original appearance and explicit examination of prior work. The percentage must not be extrapolated to all systems or generation types. Substantial similarity does not, by itself, permit inferring a psychological intention to defraud. [R10]

Read together, these results do not impose a single verdict on “AI creativity.” They show that generation, perceived originality, retrieval of prior work and allocation of credit must be evaluated separately. A book about novelty detection would be wrong to use generation success as evidence of a good detector or a case of borrowing as proof that every generated idea is derivative in the same sense.

7.2. Novelty Benchmarks and the Ground-Truth Problem

RINoBench, presented by Schopf and Färber in 2026, contains 1,381 research ideas and tools for evaluating scores and justifications. The consulted version is listed as accepted at LREC 2026. The authors report that resemblance between model-produced justifications and reference reasoning does not ensure agreement on scores. Dataset construction uses OpenReview evaluations, filtering of certain disagreements and LLM processing of ideas and justifications. The original dimensions combine novelty with technical or empirical significance. These choices matter for interpreting the measured construct. [R11]

NovBench, by Wu and colleagues, contains 1,684 paper-review pairs and is associated with ACL 2026 in the metadata of the consulted version. It includes novelty descriptions extracted from introductions and expert evaluations. The framework examines relevance, correctness, coverage and clarity of generated evaluations. The authors report limitations in the tested models' understanding of novelty and instruction-following difficulties in some fine-tuned models. Interpretation must take into account that an introduction does not necessarily contain all the details needed to compare the contribution. [R12]

Liu and colleagues propose an approach combining conceptual descriptions of contributions, retrieval of prior work and evaluation by language models. On their novelty-detection benchmarks, they report improvements from idea-adapted retrieval. The result is evidence of practical possibility in a specific configuration, not universal certification of originality. It supports evaluating the entire access-and-comparison procedure instead of assuming that the model must decide from parametric memory alone. [R13]

The general problem with any such benchmark is separating the available label from the target property. A review can combine originality with importance, clarity or credibility. A score may represent an expert's impression after limited reading, not an exhaustive search for prior work. A dataset can remain very useful even if it does not measure absolute historical priority, provided this limit is explicit.

Another risk arises when the evaluated object is itself produced through automated reformulation. If one model extracts the contribution from the paper, another model judges the extracted representation. A failure may originate in extraction, while success may reflect conformity to the extractor’s style. This is a general methodological observation, not proof that any particular benchmark is invalid. Estimating the effect requires comparisons between originals, human extractions and automated extractions, retaining the same cases.

Filtering disagreement can improve label consistency but change the population studied. If we eliminate cases on which experts disagree, we evaluate a more consensual subset. We do not automatically know how the system would perform on ideas challenging a framework or requiring unusual expertise. A benchmark can have a consensual layer for stable comparisons and a contested layer for studying limits. Both are informative, but they answer different questions.

Evaluator circularity must also be avoided. If an LLM produces the reference justification, another LLM scores the candidate justification and the main criterion is semantic resemblance between them, the result may reflect a shared stylistic culture. This risk does not prohibit automated evaluation. It calls for independent checks: concrete prior work, verifiable differences, expert-analyzed subsets and measurement of the automated evaluator’s own errors.

A strong benchmark should publish not just scores but each case’s genealogy: the original document, reference time, transformations, criterion, decisive prior work and reasons for disagreement. Without this genealogy, scores are hard to interpret when models differ. A system may perform worse because it detects a real ambiguity concealed by the single label.

7.3. Related Evidence and the Limits of Transfer between Domains

Research on detecting out-of-distribution observations provides a relevant warning, although it is not identical to our problem. Nalisnick and colleagues show that certain generative models can assign high likelihood to data from a distribution different from the training distribution. Ren and colleagues investigate likelihood ratios to control

effects of background statistics. These results weaken the naive identification of probability density with semantic membership. They do not turn out-of-distribution detection into a detector of scientific contributions. [R17, R21]

The conceptual difference matters. An idea can be new and formulated in perfectly ordinary language, thus not appearing outside the linguistic distribution. A text can be out of distribution because of language or formatting while describing an old idea. A system detecting anomalies may help route cases for verification, but cannot automatically call an anomaly a “discovery.”

Uzzi and colleagues analyzed reference combinations in scientific literature and observed an association between citation impact and combinations blending a conventional foundation with atypical elements. The study offers a perspective on the relationship between familiarity and the unusual in science. It does not prove that a particular combination is true, original or causally responsible for impact. Citations are not identical to epistemic value. [R22]

For a detector’s architecture, the lesson is not to mechanically reward rare combinations. A combination may be rare because it is unsuitable. Nor should we assume that a new contribution must be strange in every component. A new mechanism may need conventional presentation to be verifiable. Tests must allow a mixture of familiar foundation and local difference, not just an artificial opposition between banality and extravagance.

Shumailov and colleagues studied model degradation under recursive training on generated data and the loss of aspects of the original distribution. Gerstgrasser and colleagues show that accumulating real and synthetic data can avoid certain forms of collapse observed when data are replaced. The results do not authorize the conclusion that every use of synthetic data destroys diversity. They require specifying the data-recycling mechanism and continued access to external observations. [R23, R24]

In novelty evaluation, an analogous scenario can be formulated but must be tested separately. If only accepted ideas return as positive examples while rejected ones disappear, the visible distribution may narrow. However, the effect depends on selection policy, external

ideas and archive preservation. It must not be presented as an inevitable consequence of the language model. It is a hypothesis about the decision ecosystem.

The balance of the literature is therefore asymmetric. There is evidence that LLM judges can have biases, that novelty evaluation remains difficult and that clear explanations do not guarantee correct comparisons. There are also results showing the usefulness of assisted generation and retrieval. Neither group demonstrates a universal thesis about all models and criteria. What the literature justifies is a more demanding validation program, not a metaphysical verdict.

Chapter 8. How to Identify a Familiarity Bias Experimentally

8.1. Hypotheses, Interventions and Alternative Explanations

A good experiment begins by defining the claim that could be wrong. “Models prefer the familiar” is too broad unless we specify familiarity and the task. We can define a behavioral hypothesis: for the same contribution and prior work, conventional wording receives a higher novelty score than unusual wording under conditions of comparable clarity. We can define an informational hypothesis: a particular compression stage reduces recovery of the decisive semantic difference. We can define a causal hypothesis about training: increasing exposure to a contribution pattern, with other factors controlled, changes its evaluation independently of genuine novelty.

These hypotheses require different experiments. Reformulating the prompt does not intervene on pretraining. Comparing two commercial models does not isolate data frequency because the models may differ in architecture, capacity, post-training and tools. Observing a correlation between perplexity and score does not separate clarity, correctness and familiarity. A study must declare the level at which it claims a conclusion.

We can informally represent a causal mechanism through variables. F is training exposure, S style, N reference novelty, D semantic difficulty, R access to prior work, Z the intermediate representation and J the verdict. F can influence both Z and J. S can influence difficulty and perceived quality. N is not necessarily observed directly; sometimes we have only an expert label Y influenced by other factors as well. An experiment ignoring these relationships can attribute to novelty what belongs to style or difficulty.

Factorial design allows factors to vary independently. We can combine old and new contributions with familiar and unusual formulations, then with prior work absent, automatically retrieved or supplied by

an experimental oracle. In another block, we can compare full text with a summary. We need not run every combination immediately. But we must understand what interactions might change the result. A style effect may appear only when prior work is absent; that is a different problem from a bias persisting with complete information.

The experimental oracle is not an omniscient agent. It is the condition in which the experimenter supplies exactly the decisive prior work and the information needed for comparison. In a finite world, this information can be checked exhaustively. In a real study, “oracle” must be used more modestly: an expert-adjudicated package of prior work. The condition’s role is to separate search failure from comparison failure.

A simple causal contrast is the difference between scores for the same idea under two styles, with random assignment of order and instances. For idea i , we can denote potential outcomes $J_i(1)$ and $J_i(0)$, where 1 and 0 indicate the stylistic treatment. The target average effect is the mean difference. The definition does not automatically resolve identification: memory effects between evaluations must be avoided and content equivalence ensured.

$$\tau_S = E[J_i(1) - J_i(0)].$$

In practice, the same conversation session can create dependencies between conditions. If the model has already seen the original version, evaluation of the paraphrase is no longer independent. We can use separate sessions and randomization. If an interface retains memory or dynamically changes the model, these conditions must be documented. A repeatable experiment needs the version, date, parameters and complete input–output log.

Hypotheses must be accompanied by criteria for local refutation. If the style effect is smaller than a preregistered practical threshold with sufficiently small uncertainty, the hypothesis of a relevant effect in that configuration is weakened. A nonsignificant result in a small sample is not a refutation. Likewise, a significant but tiny effect may have little importance for selection. Effect size and decision cost must be discussed together.

8.2. Measurement: From Scores to Recovered Contributions

Overall accuracy is insufficient when novelties are rare. A classifier calling every idea “old” can have high accuracy and no usefulness for discovery. Saito and Rehmsmeier discuss why precision–recall curves are often more informative than ROC curves for evaluation on imbalanced classes. In our application, metric choice must be explicitly linked to the filtering objective. [R19]

Recall measures the proportion of genuinely new contributions retained. Precision measures the proportion of selections that are genuinely new. They express different costs. For a team able to examine twenty proposals a week, the practical question may be how many new contributions are in the top twenty. For an alert system, missing almost no cases may matter even with many false alarms. There is no universally optimal threshold.

An appropriate metric is recall at a fixed budget. For each method, we allow the same number of candidates to be sent to an expert. We compare how many novelties are recovered. We can add the total cost of tokens, searches and expert time. A system detecting five percent more contributions but consuming ten times more verification may be better or worse depending on the application. Cost must not be hidden behind a single curve.

The quality of the identified difference must be measured separately. The model can correctly select a candidate for a false reason. We can evaluate whether the cited passage exists, whether the prior work is relevant, whether the anticipation relation is valid and whether the difference witness works. A correct selection with a false explanation is not equivalent to an evidence-supported selection, especially when other systems will reuse the result.

Abstention must be handled explicitly. An evaluator may answer “insufficient information.” If we count every abstention as an error, we favor systems that guess. If we exclude it completely, a system can abstain on every difficult case and report artificially high precision. Coverage, risk on decided cases and the cost of escalated cases must be reported together. A good selective system allocates its uncertainty usefully rather than hiding it.

Calibration can be evaluated by comparing probabilities with observed frequencies. But good average calibration can conceal large errors on rare contributions, underrepresented languages or ideas changing assumptions. Analysis of subgroups defined before the test is necessary. We should not create dozens of subgroups after seeing the results and report only those supporting the preferred hypothesis.

Conformal prediction methods can offer marginal guarantees under assumptions such as data exchangeability. The introduction by Angelopoulos and Bates explains these mechanisms and limits. In a stream of discoveries, the distribution may change precisely because new contributions appear. A guarantee calibrated on the past must therefore not be presented as individual certification of historical novelty or automatic protection under distribution shift. [R20]

The statistical unit must be chosen correctly. Ten paraphrases of the same idea are not ten independent discoveries. A hundred runs of the same model on the same pair are not a hundred replications of the training effect. The data have hierarchical structure: problem families, ideas, variants, models and seeds. Confidence intervals must reflect this dependence through idea-level aggregation, cluster bootstrap or suitable hierarchical models.

One possible statistical model, presented here as a planning tool, uses logistic regression for verdict correctness, with effects of novelty, style, access and compression, plus preregistered interactions and random intercepts for family and idea. The model does not turn observation into causality. Causal interpretation comes from the intervention design; regression estimates contrasts and uncertainty under its assumptions.

8.3. Testing Training Frequency and Statistical Power

Testing the initial hypothesis's exact explanation requires control of exposure. We can construct an artificial domain where certain contribution forms are presented as innovative more often and others less often. In the test, novelty is determined independently of these fre-

quencies. If evaluations follow the frequency of the “innovative” label despite the controlled truth, we have evidence for the proposed causal mechanism in that regime.

Two manipulations must be separated. The first changes the frequency of an idea’s content. The second changes how frequently an idea form is associated with an assessment of novelty. These can have opposite effects. Repeating an idea can help the model recognize it as old. Repeating praise for a pattern can make the model regard it as innovative. The initial hypothesis is more interesting in the second form because it concerns learning an evaluative concept, not merely memorizing content.

Total token count, optimization steps, difficulty distribution and label proportions must be controlled. Otherwise, a higher-exposure condition may simply receive more useful training. We can redistribute the same resources among categories and use multiple independent initializations. Comparing a single pair of models does not allow the effect to be robustly separated from optimization variation.

Small models or local adaptations can provide an accessible causal laboratory. A result in such a laboratory must not automatically be transferred to the pretraining of a very large model. It can demonstrate the mechanism’s existence and the conditions in which it appears. The next step is testing transfer to different capacities, objectives and data regimes. This progression is stronger than attempting to deduce an opaque system’s training data from a few answers.

Sample planning must start from the minimum relevant effect. For a paired contrast between two styles, information comes especially from cases where verdicts differ. If almost all pairs produce the same verdict, a larger sample may be needed to estimate a small difference. For ordinal scores, we must decide whether distances between grades are treated as equal or whether an ordinal model is used. The choice must not be made after seeing which analysis produces the desired result.

A power simulation is useful if the generating assumptions are explicit. We can assume particular error rates and correlations between paraphrases, generate synthetic data and estimate how often the procedure would detect an effect. These results concern statistical design

under the chosen assumptions, not actual LLM performance. The book does not turn such planning simulations into empirical evidence for the familiarity hypothesis.

Preregistration must fix the primary outcome, contrasts, relevance threshold, exclusions, handling of abstentions and rules for exploratory analyses. It should not prohibit unexpected discoveries. It should separate confirmation of a prior prediction from formulation of a new one after observing the data. An interesting finding discovered later remains valuable but needs independent replication before being presented as confirmation.

Ultimately, the hypothesis should be evaluated through a profile of results. A style effect without a controlled-exposure effect supports a behavioral bias, not its training explanation. A compression effect that disappears with access to the original supports a remediable informational mechanism. Persistent failure with perfect prior work and a simple criterion indicates a comparison limit in the tested configuration. Robust success in the finite world refutes universal impossibility without automatically certifying competence in open science.

Chapter 9. The Controlled Laboratory: Novelty with Exact Ground Truth

9.1. Why Finite Circuits Are a Suitable Laboratory

To test a detector, we must know what it should have detected. In open science, this condition is difficult: prior work may be missing, experts may have different criteria and the contribution may not yet be validated. A finite laboratory does not replace reality, but it allows certain mechanisms to be isolated. Boolean circuits are suitable because they have exact semantics, can be reformulated in many ways and can differ on a single case.

A Boolean circuit takes n binary inputs and produces a binary output. AND, OR, NOT and XOR gates have determinate semantics. For fixed n , the circuit's function can be represented by a table with 2 to the power n rows. If two circuits have the same table, they are extensionally equivalent on that domain. If the tables differ, at least one input witnesses the difference.

We define a library K of circuits and a candidate c . Relative functional novelty is one if c 's function coincides with none of the functions of elements in K . This definition is deliberately narrow. It does not say that c has an original implementation, is more efficient or was discovered for the first time in history. It says that the library contains no identical function. The advantage is that the verdict can be calculated exactly, independently of a linguistic evaluator.

A minimal example uses two inputs a and b . The library contains AND and OR. The candidate "negation of the disjunction of the negations" is equivalent to AND and must be classified as old despite different wording. Candidate XOR differs from both and must be classified as new relative to this library. If we add XOR to the library, the same candidate becomes old. We thus test both invariance to reformulation and correct dependence on the reference.

This example is too easy for a broad evaluation because gate names may trigger memorized knowledge. We can define operations with random names and supply their tables locally. We can construct expressions with different trees, redundant operations and variable permutations. We can require the model to use only the definitions in context. The aim is not to conceal meaning artificially but to eliminate the possibility of answering solely from a name’s reputation.

The verifier and generator must remain separate. The generator creates candidates and variants. The verifier calculates tables and labels. The LLM produces the verdict or translates the description into a verifiable representation. If the same model generates the labels and is then evaluated against them, the laboratory loses the very independence that makes it useful. In the executable appendix, labels are calculated by exhaustive enumeration without calling a model.

A finite laboratory can also test operational novelty, but the criterion must change. Two equivalent circuits can have different depths or gate counts. If the new candidate is smaller than every implementation in the library, we have another type of relative novelty. It does not follow that it is the globally minimal circuit; that would require exploring a larger space or proving a bound. Every extension of the criterion brings a verification obligation.

9.2. Rare Differences, Collisions and Test Families

A critical test of the initial hypothesis must include rare differences. We choose a function f and construct g to coincide with f on every input except one. If inputs are uniform, the difference appears with probability 2 to the power minus n . For n equal to eight, only one row out of 256 distinguishes the functions. Functional novelty is exact, but a small sample of behaviors can miss it.

If we observe m independent inputs with replacement, the probability of missing the difference entirely is as follows:

$$P(\text{missed difference}) = (1 - 2^{-n})^m.$$

The formula concerns the information supplied by sampling, not LLMs. To reduce the probability of missing the difference below α , the budget must satisfy the relation derived by taking logarithms. This

analysis shows that failure can result from the absence of a distinguishing case even if the evaluator perfectly compares every observation received.

$$m \geq \frac{\log \alpha}{\log(1-2^{-n})}.$$

We can compare three conditions. In the first, the model sees only a sample without the witness. In the second, it sees a sample containing it. In the third, it receives complete expressions and can reason symbolically. The first condition does not permit reliable identification of the difference from examples alone; the second provides local evidence; the third tests the ability to deduce the difference from structure. These conditions must not be aggregated into a single success rate.

To test smoothing, we can construct a representation that deliberately removes the distinguishing row or retains only the circuit’s thematic category. The collision is then known by construction. An evaluator based exclusively on the compressed representation cannot distinguish the pair. This experiment checks the pipeline’s informational limit, not a particular LLM’s inability. We can subsequently replace the deliberate compressor with a linguistic summarizer and measure how often a relevant collision occurs.

Another family uses reformulations without changing the function. We can apply double negation, De Morgan’s laws, reordering of commutative arguments and introduction of redundant subexpressions. These transformations produce different texts with the same label. If the model considers them new only because the expression is long or unusual, we have a false positive for syntactic originality. If it correctly normalizes all expressions, we have useful semantic-invariance competence.

The complementary family changes a single relevant component while keeping the form as close as possible. Replacing an OR gate with XOR can produce a local functional difference. Changing the input domain can turn equivalence into nonequivalence. Moving a quantifier in a description can change the objective. These minimal pairs are useful because they reduce the chance that the model uses length or topic as the main cue.

One danger of synthetic datasets is the emergence of artifacts. If every new example has an extra level of parentheses, the model can learn the parentheses. If every old example is obtained through a single rewriting rule, it may recognize the rule without general comparison. The generator must balance length, structure and vocabulary between classes. More importantly, the test must be split by latent transformation families, not merely by randomly dividing nearly identical examples.

We can reserve a rewriting family exclusively for testing, a set of base functions exclusively for testing and new names for each round. This does not guarantee absence of every shortcut, but it makes generalization more demanding. Results must be reported for each family. A good average can conceal failure precisely on the rare differences or quantification changes motivating the research.

9.3. What a Positive or Negative Result Would Demonstrate

Suppose a model correctly classifies new circuits, recognizes old reformulations and produces valid difference witnesses on families not used in preparing the test. The result would demonstrate comparison ability in a finite domain under the experimental conditions. It would strongly weaken the thesis that every recognition of novelty is merely retrieval of memorized ideas. It would not demonstrate that the model can examine the open scientific literature or assess an idea's importance.

If the model succeeds only when given complete tables but not descriptions, the problem seems connected to interpretation or symbolic computation. If it succeeds when the witness is supplied but cannot find it, we separate verification from witness search. If it fails even when the difference is explicitly highlighted, we can investigate the instruction, context or ability to apply the criterion. These results are more informative than simply saying "it does not understand."

If all models fail after compression has removed the witness and structural information, we do not have a useful comparison between models. We have constructed an unidentifiable task. The experiment's

value is to demonstrate the architectural defect. A good detector should at least recognize insufficient information and abstain. Certainty in a verdict under this condition is itself a calibration result.

If style affects the verdict even though the circuit and library are identical, there is a behavioral effect. Interpreting it requires checking that the variants describe exactly the same circuit. We can generate descriptions from a controlled grammar and then include validated human variants. Increasing linguistic realism reduces formal control; a progression from controlled language to natural explanations is therefore useful, rather than abruptly replacing one with the other.

A negative laboratory result does not prove that the model will fail on every real task. Humans can use analogies and domain knowledge poorly represented by random circuits. But success on subjectively assessed real tasks does not cancel formal failure either. The two tests measure different components. A good research program preserves the tension between control and relevance instead of choosing only the favorable test.

For reporting, each case should include the expression or description, library, criterion, exact label, existing witnesses, stylistic transformations and the evaluator's raw output. If the model proposes an intermediate circuit, it must be retained separately from the reference circuit. A correct answer obtained from an incorrect translation may be accidental. An incorrect answer from a correct translation indicates another problem.

The laboratory can be extended to finite automata, regular expressions on bounded domains, finite Datalog relations or discrete causal models. Each domain offers different distinctions: temporal behavior, quantification, dependencies or interventions. Extensions should not be made merely to add benchmarks. They must test a precise hypothesis about what kind of difference is lost or recognized.

The laboratory's ultimate aim is not to proclaim a winner. It is to construct an anatomy of error. A system missing prior work needs a different remedy from one that reads it and erases the difference. A system that can verify a witness but not search for one has a different profile from one that confuses the criterion. Only after this anatomy is established is scaling to real documents worth discussing.

Chapter 10. Validation on Real Research

10.1. Temporal Reconstruction and the Limits of Retrospective Tests

A test on real papers has obvious appeal: the ideas are authentic, the language rich and the difficulties resemble those encountered in use. But document authenticity does not guarantee label validity. Evaluating novelty at publication requires reconstructing what was available beforehand, what the paper claimed and under what criterion it was considered a contribution. A contemporary database can easily mix documents from before and after the event being evaluated.

The first principle is temporal freezing of the reference corpus. Every source must have a verifiable version and date. The preprint, final article and updated documentation may have different content. If a decisive passage was added after the reference date, it must not be used as evidence that the idea was already known. Likewise, a later article retrospectively explaining the contribution cannot be treated as prior work.

The second principle is recognizing the limits of anonymization. Removing names and the title does not necessarily erase a famous paper's identity. Its formula, dataset or argument structure may allow recognition. A model trained after publication may know the historical outcome of the evaluation. In such a test, we cannot automatically attribute success to local comparison. We must report possible contamination and use analyses that reduce it.

One solution is to combine a retrospective study with a prospective one. In the prospective study, the model and procedure are fixed before candidates are collected. Contributions are evaluated before stable public consensus exists. Subsequently, discovered prior work, validations and revisions are tracked. This design reduces some forms of contamination but costs time and does not immediately produce final truth. Real research can remain controversial for years.

Prospective candidates must not be selected only from ideas that have already passed an LLM filter. Otherwise, the test population is conditioned on the suspected mechanism. An independent sample is needed, possibly stratified by domain, contribution type and maturity. Rejected or unclear ideas must be retained in a separate layer rather than eliminated without a trace. Otherwise, the very cases that could confirm the hypothesis disappear before evaluation.

A realistic protocol can distinguish claimed novelty, novelty supported by comparison with prior work and novelty remaining after technical validation. These states can change over time. A proposal may seem new and then be anticipated by a source discovered later. It may seem wrong and later receive a proof. The system must allow the verdict to be revised without falsifying its history. Evaluated novelty is a versioned object, not a property fixed forever at first reading.

10.2. Experts, Disagreement and Verifiable Labels

Experts are indispensable in open domains, but they are not perfect oracles. A reviewer may know one subfield very well yet miss a contribution from a neighboring area. They may interpret “new” as “important,” penalize lack of validation or use a different definition of the problem. These possibilities must not be used to relativize every criticism. They justify a protocol requiring reasons and sources, not just grades.

A useful expert evaluation begins by extracting the contribution. The reviewer must specify exactly what is being compared. They then identify relevant prior work and describe the relation to each item. Finally, they separate the verdict on the difference from the verdict on correctness and value. A justification such as “the idea is incremental” must be translated into a concrete relation: what already existed and what remains. Without this translation, agreement between evaluators may be merely agreement on an impression.

Disagreement must be classified. Sometimes experts disagree about the text’s meaning. Sometimes about the existence of prior work. Sometimes about the difference’s relevance. Sometimes about technic-

al validity. These disagreements require different remedies: clarification, further search, specifying the criterion or experiment. A single agreement coefficient does not locate the problem.

We can preserve multiple labels and a reasoned adjudication. Adjudication must not erase the initial positions. If newly found prior work changes the verdict, the log must show the change. For some cases, the correct label may remain “undetermined under current conditions.” These cases are not statistical rubbish. They matter for evaluating abstention and understanding the system’s limits.

A minimal rubric can be organized in two columns without reducing evaluation to a mechanical form.

Dimension	The Question to Resolve
The Contribution	What claim, mechanism or capability does the text claim to introduce?
The Prior Work	Which earlier source already contains the relevant element, and in which passage?
The Difference	What remains different after comparing assumptions, results and resources?
Validity	What evidence supports or refutes the claimed difference?
Uncertainty	What missing information could change the verdict?

Evaluators must have access to comparable resources. If a human can search for two hours while the model receives only a summary, the comparison also measures access. If the model can consult a very large index while the human cannot, the situation is reversed. Two evaluation levels are useful: comparison competence with the same source package, and end-to-end performance with each system’s own tools at declared costs.

Texts in different languages deserve a separate layer. Translation can improve accessibility but change terminology or smooth differences. A multilingual study must retain the original, translation and any am-

biguities. It is not enough to translate all documents with a single LLM and assume that linguistic difficulty has been eliminated. It may merely have moved into an invisible stage.

In disputed situations, concrete prior work carries a different weight from a similarity score. If the evaluator shows a passage that exactly anticipates the contribution, we have verifiable evidence. If it merely says that the idea “resembles” a domain, we have an investigative lead. The system must distinguish these evidence levels. Otherwise, a vague association can be presented with the authority of a refutation.

10.3. A Longitudinal Study That Can Change the Conclusions

A realistic program can begin with a small, well-documented retrospective corpus to check the annotation procedure and tools. The next stage introduces new proposals, blindly evaluated by models and experts. The longitudinal stage follows what happens to them: prior work appears, they are proved, falsified or remain unclear. These stages have different roles and must not be combined into a single success score.

The primary outcome could be recovery of contributions later confirmed, at a fixed initial verification budget. But “later confirmed” must be carefully defined. Publication is not identical to truth, citation is not identical to novelty and lack of follow-up may reflect lack of resources. A longitudinal evaluation must track evidence and comparisons, not merely an idea’s social success.

To avoid circularity, experts assigning later labels should not know the model’s initial score, insofar as this is practically possible. If the score influenced funding or attention to the idea, the later outcome is no longer independent of selection. A favored idea may receive more development and subsequently appear more valuable. This is an effect of intervening on resources, not necessarily a correct prediction of a preexisting property.

We can reduce the problem through an audit sample funded or evaluated independently of the score. Some rejected ideas are randomly selected for in-depth examination. This procedure allows false negatives to be estimated. Without it, we know in detail what happened to ac-

cepted ideas and almost nothing about those eliminated. A filter may look excellent precisely because it has made its errors impossible to observe.

Costs must be recorded by component. How long does contribution extraction take? How many searches are needed for prior work? How much time does verifying a false witness consume? What does clarifying an ambiguity cost? A model producing many apparently precise but hard-to-check justifications may transfer costs to the expert. Efficiency is not merely time to the first verdict but time to a supported, auditable conclusion.

Negative results are informative when localized. If a contribution class is repeatedly missed, we must determine whether its prior work is hard to retrieve, it changes the ontology or it contains small logical differences. Calling it “radical novelty” after the model fails is insufficient. The category must be defined through properties independent of the outcome. Otherwise, the hypothesis becomes tautological: we call radical what the system does not recognize and then conclude that the system does not recognize the radical.

At the end of such a study, the conclusion may be mixed. The system may be very good at identifying close rediscoveries, useful for mapping prior work and poor at evaluating framework changes. It may save time without being suitable for automatic rejection. Such a result is not a partial victory to conceal. It is exactly the operational profile a community needs to use technology without attributing undemonstrated capabilities to it.

Chapter 11. The Architecture of an Evidence-Based Evaluator

11.1. Preserving the Evaluated Object and Separating Operations

The architecture proposed in this chapter is not an implemented, validated product. It is a research contract for a system using LLMs without turning their verdict into opaque authority. The central idea is that the model must construct verifiable objects: claims located in text, relations to prior work, explicit differences and questions that tools or experts can resolve.

The system's input must be retained as an immutable or versioned artifact. The original document, date, provenance and any transformations are needed for auditing. A hash can help identify the version but does not demonstrate the truth of its content or the fidelity of its interpretation. Cryptography can certify that we are examining the same bytes; it does not certify that we understand them correctly. This separation matters to avoid confusing technical integrity with epistemic validity.

The first operation extracts claimed contributions. Every contribution must be linked to passages and a degree of explicitness. Some claims are directly stated; others are inferred by the evaluator. The categories must be marked differently. If the system invents a guarantee to make the proposal more coherent, it changes the evaluated object. A charitable interpretation may be retained as an alternative hypothesis, not silently substituted for the original text.

The second operation constructs comparison questions. For each contribution, the system searches for prior work concerning mechanism, result, assumptions and resources. It must not assume that a single query captures all these dimensions. A contribution can have the same mechanism with a different guarantee, or the same guarantee with different assumptions. Search paths must reflect this structure and preserve unusual terms that may be decisive.

The third operation compares retrieved sources with the candidate. The result is not yet a global score. It is a map of relations: the same claim, a narrower result, a different assumption, merely thematic prior work, contradiction or insufficient information. Relations must be accompanied by evidence. A relevant passage in a paper cannot be replaced by simply citing the paper's title.

The fourth operation attempts to verify differences. For programs, it may search for distinguishing inputs. For a proof, it may check a formalizable step. For an empirical proposal, it may identify the experiment that would separate its mechanism from prior work. Not every difference can be certified automatically. But the system can specify what kind of evidence would be needed and which part of the verdict remains conditional.

Only then comes aggregation. A score, if needed for prioritization, must be derived from relations and an explicit policy. Users must be able to see whether a low score comes from strong prior work or missing information. These situations should not receive the same operational label. "Anticipated" and "undetermined" require different actions.

An output format can be described through a few fields without imposing an exhaustive ontology.

Field	Required Verifiable Content
Contribution	The evaluated claim and the original passages supporting it.
Reference	The corpus, date and criterion under which novelty is judged.
Prior Work	Related sources, relevant passages and search limitations.
Difference	What remains unanticipated and under which interpretation.
Witness	Example, calculation, counterexample or verification procedure.
Verdict	Conditional decision, uncertainty and the reason for any abstention.

This contract can be implemented as a graph of operations with explicit inputs and outputs. Each node consumes versioned artifacts and produces a result with provenance. If a source changes or new prior work appears, only dependent conclusions need be invalidated. We need not claim that the LLM’s entire internal reasoning is transparent. We can demand transparency about the data, transformations and external evidence invoked by the verdict.

11.2. Diverse Retrieval, Precise Comparison and Return to the Source

Retrieval-augmented generation architectures combine information in the model with documents retrieved at use time. Lewis and colleagues demonstrated the usefulness of such a combination in knowledge-intensive language tasks. This idea offers a principle for access to prior work, not a guarantee that retrieved sources are complete or the novelty relation is evaluated correctly. [R14]

A lexical index is good for exact terms, formulas, acronyms and rare expressions. A semantic index can retrieve related descriptions worded differently. Multivector representations can preserve finer interactions between parts of text. ColBERT, by Khattab and Zaharia, is an example of retrieval with late interaction between token representations. Using such a technique does not mean that novelty differences are automatically preserved; that must be evaluated on the concrete task. [R15]

The proposed strategy is justified retrieval diversity. We search for both the topic and the removed assumption, both the mechanism and the distinguishing consequence. We retain results that are close on different dimensions. A global top-k can eliminate important prior work if it is lexically distant but semantically decisive. We can allocate separate budgets to search paths and then deduplicate, without assuming that a single similarity score suffices.

The final comparison must use passages complete enough to include the conditions. An abstract may state a guarantee without listing every assumption. A sentence in the conclusion may be broader than the demonstrated result. The system must be able to follow internal

references and check where the claim is justified. This capability can be costly; separating initial filtering from in-depth examination is therefore useful.

Summaries must be treated as views, not permanently authoritative replacements for documents. A summary is useful for navigation and identifying questions. When the verdict depends on an exception, negation or quantifier, the system must return to the source. An escalation rule can require original access when two possible interpretations would change the decision. This rule can be tested through minimal pairs.

A comparator can produce an internal table of assumptions and guarantees, but must allow the value “unspecified.” Automatically filling gaps with typical assumptions can create false equivalence. If prior work does not state a property in the available passages, we cannot assume either that it has or lacks that property. Relevant context must be sought or uncertainty marked. Absence of a claim from an excerpt is not identical to its negation in the full paper.

An adversarial verifier can attempt to refute each proposed relation. If the model says A anticipates B, the verifier searches for a difference. If it says B is new, the verifier searches for prior work or an equivalent interpretation. The roles must produce evidence, not just opinions. Agent count is no substitute for evidence independence. Three agents using the same mistaken interpretation can reinforce the error.

The architecture must also handle documents as potentially hostile data. An article may contain instructions or text attempting to influence the evaluator. These must not be confused with application instructions. Separating analyzed content from tool control is an engineering requirement. In a research system, compromising this separation can affect not just security but experimental validity: the model is no longer answering the same task.

11.3. Uncertainty, Audit and Criteria for Accepting the Architecture

An evidence-based system must represent uncertainty by component. It may be certain that two formal expressions differ, uncertain that translating the text into expressions was faithful and very uncertain

that the library covers all prior work. A single confidence percentage conceals this structure. It is more useful to distinguish uncertainty of interpretation, retrieval, comparison and validation.

Not all these uncertainties need be well-calibrated probabilities from the outset. They can be operational states accompanied by reasons: insufficient sources, alternative interpretations, an unverified witness or an incomplete corpus. What matters is that labels lead to appropriate actions. Missing prior work may call for further search; an ambiguous translation may call for author clarification; a verified formal difference may proceed to value assessment.

An audit must preserve the decision path. Which model version was used? Which prompt? Which documents were accessed? Which queries were executed? Which passages supported the conclusion? Which operations were repeated? This information enables reproduction and diagnosis. It must not be confused with guaranteed exact reproduction of an external service that may change. The report must specify what can be rerun and what depends on the provider's infrastructure.

Architecture acceptance criteria must be formulated before full implementation. For example, in the finite laboratory, semantically equivalent transformations should not change the verdict beyond the preregistered threshold. In cases with supplied prior work, anticipation relations must be supported by correct passages. On deliberately unidentifiable cases, the system must abstain or formulate a conditional verdict. On the real stream, rejection audits must estimate the rate of lost contributions.

The architecture would be refuted in its current form if it adds substantial cost without reducing decisive errors, if evidence is frequently fictitious or if access to the original does not correct compression-induced losses. It would need revision if the ambiguity detector escalates almost every case or if witness verification costs more than direct human evaluation. A technical proposal does not become good merely by being more complex than a simple prompt.

A minimal architecture might actually be better: two independent contribution extractions, diverse search, passage-level comparison and output separating prior work from uncertainty. Additional com-

plexity must be justified through ablation experiments. If removing an agent does not change results, its role may be redundant. If a simple quantifier-preservation rule resolves many errors, it may be worth more than an expensive evaluator ensemble.

The final principle is that the system must be correctable. A new source must be able to change the verdict. An interpretation error must be locatable. A lost difference must be recoverable through access to the original. These properties do not guarantee discovery of every new idea, but reduce the risk that a fluent, familiar verdict ends the investigation.

Chapter 12. Filtering Rare Ideas as a Decision Problem

12.1. Recognizing Is Not the Same as Selecting

Even a perfect detector of relative novelty does not automatically solve selection. A team may need feasible ideas, impactful results or projects compatible with its resources. Novelty is one decision component, not the sole objective. If these criteria are mixed from the outset, we can no longer determine whether the system missed originality or made a different strategic choice.

We can represent the decision through the expected utility of examining a proposal. Utility may depend on the probability that the difference is real, the possible value of the result, verification cost and the opportunity to learn. These components are the decision-maker's assumptions. They are not properties the LLM can objectively infer from the text. A system must request or declare the selection policy.

A simplified example can have two actions: in-depth verification or postponement. If verification costs c , a valid contribution brings value v and the estimated probability of relevant validity is p , the rule $p \cdot v > c$ can justify verification in the minimal model. But v may be highly uncertain, and verification can produce useful information even if the idea is false. The example shows the decision's structure, not a universal research threshold.

Very rare novelty complicates estimation. If we have few confirmed examples, probabilities are hard to calibrate and model errors may dominate. A cautious policy can combine exploitation of scores with exploration of unusual cases. We need not randomly select half the projects. But a system that never explores beyond its confidence region will not systematically learn where it is wrong.

Semantic distance should not be used as a direct substitute for exploration value. A very distant text may be noise. A small difference in a central mechanism may matter much more. Exploration can be stratified by uncertainty type: hard-to-find prior work, assumption

changes, contributions in different languages or cases where models and verifiers disagree. The choice of strata is a hypothesis that must be evaluated through auditing.

There is also the negative value of a false verdict that something is old. If the system says “this already exists” without concrete prior work, the author may abandon a promising direction. In a brainstorming application, this cost may exceed that of a false novelty alarm. In an editorial application, costs may be distributed differently. Response policy must reflect context, not merely average accuracy.

A normative recommendation of this book is that research systems should be more demanding when rejecting than when proposing an investigation. Claiming “worth checking” requires a plausible lead. Claiming “there is nothing new” requires a comparison sufficient to eliminate the claimed differences. This asymmetry is not a statistical law; it is a choice justified by the irreversible cost of eliminating rare ideas.

12.2. How We Observe Errors Concerning Rejected Ideas

A filter produces two populations: accepted and rejected candidates. Usually, accepted candidates are checked more carefully. We thus learn much about false alarms but little about missed novelties. If we report only selection precision, we may perceive an excellent system while recall remains unknown. This observational asymmetry must be corrected through design.

One solution is random sampling of rejections for auditing. Every rejected candidate receives a known, nonzero probability of being checked. Probabilities may differ between strata, for example to oversample highly uncertain cases. But they must be retained for weighting estimates. If we check only rejections that seem interesting retrospectively, we do not obtain a representative estimate.

Let A_i indicate that idea i was audited, q_i its inclusion probability and Y_i the novelty label established through auditing. The total number of novelties in the rejected population can be estimated using the following weighted sum. The justification is elementary: the mean of A_i/q_i is one if inclusion follows the stated probability.

$$\hat{T}_{\text{missed}} = \sum_{i \in \text{rejected}} \frac{A_i Y_i}{q_i}.$$

The estimate can have high variance when audit probabilities are very small. A formally nonzero chance is not enough. The budget must allow useful uncertainty intervals. If novelties are extremely rare, auditing may require extensive examination. This is a property of the problem, not an error in the formula. An adaptive strategy can increase sampling in strata where signals appear while retaining appropriate estimation rules.

The audit must also track the reason for rejection. A system can miss contributions because it considers them old, unclear, infeasible or insufficiently supported. If the policy permits rejection for infeasibility, cases where novelty was recognized must be separated. Otherwise, a false-negative analysis may attribute to the detector a decision made by another component.

Another question is who assigns the label after auditing. If the same LLM and sources are used, the procedure may reproduce the initial error. Auditing must bring additional information or verification: experts, broader search, formal tools or time for experimentation. Repeating the same verdict with a different prompt can test stability but is insufficient as an independent audit.

The rejection archive also has a longitudinal role. An idea considered unclear may become intelligible after a new framework appears. Earlier work may be discovered later. Preserving versions enables reevaluation and learning from errors. Deleting rejections makes selection irreversible, and irreversibility reduces the system’s ability to learn what types of novelty it fails to recognize.

12.3. Selection Dynamics and the Risk of Homogenization

We can construct a very simple model of selection’s cumulative effect. We have a distribution $p_t(x)$ of visible ideas at round t and an acceptance probability $a(x)$. If only accepted ideas return in the next round,

the following distribution is proportional to the product of visibility and acceptance. The model is a didactic idealization, not an empirical description of a community.

$$p_{t+1}(x) = \frac{a(x)p_t(x)}{\sum_u a(u)p_t(u)}.$$

If two classes have constant acceptance probabilities a_1 and a_2 , their frequency ratio is multiplied at each round by a_1/a_2 . A small but persistent advantage can become large after many rounds. This conclusion is algebraic. It does not say that the advantage is unjustified or every ecosystem works this way. It shows why a modest bias deserves dynamic evaluation as well as measurement of the score difference in a single round.

Introducing external ideas changes the process. We can mix the selected distribution with a distribution of new input r , weighted by λ . If λ is positive and r maintains certain classes, they do not automatically disappear. Archive preservation and reevaluation can also change acceptance probabilities. Homogenization is therefore not inevitable; it depends on input and correction mechanisms.

$$p_{t+1} = (1-\lambda) \text{Select}(p_t) + \lambda r.$$

In a real system, acceptance probability may depend on frequency. An idea may be favored because it is familiar or, conversely, because it is rare. If rarity is mechanically rewarded, the system may produce superficial variations to avoid resemblance. If familiarity is mechanically rewarded, it may preserve dominant patterns. The objective should not be to maximize either extreme, but to preserve the ability to verify relevant differences.

A portfolio policy can retain projects with different profiles: robust incremental contributions, ideas with clear differences but incomplete validation, and explorations challenging the criterion. They need not all receive the same resources. But evaluating them should not force them onto the same scale without explanation. A conceptual-clarification project and an operational-optimization project can have different types of outcome.

This discussion separates two normative questions. Do we want the system to imitate the community's current judgments, or help the community discover its blind spots? The first task can be evaluated through expert agreement. The second requires cases where disagreement is investigated and sometimes proves justified. A model perfectly reproducing consensus is not thereby a perfect detector of contributions that will change consensus.

Opposition to consensus should not be romanticized. Many rejected ideas are wrong or insufficiently developed. But a research system cannot learn which exceptions deserve preservation if it equates unfamiliarity with meaninglessness. Selection policy must allow verification of cases current scores do not favor, and audit results must be able to change the procedure.

The operational conclusion is that the novelty detector and attention-allocation policy must be designed together but evaluated separately. The detector identifies relations and uncertainties. The policy decides how much cost to accept for exploration and how much risk for rejection. Mixing them into a single score can conceal both epistemic biases and normative choices.

Chapter 13. The Strongest Objections in Both Directions

13.1. The Skeptical Case: The Model May Learn Innovation's Reputation, Not Innovation

The strongest version of skepticism does not say that a model cannot produce a sequence absent from training. That would be too easy to contradict. It says that the evaluative notion of novelty is learned from a community's verdicts, and those verdicts contain conventions, hierarchies and preferences. The model may become a good predictor of innovation's social recognition without becoming a good detector of differences not yet socially recognized.

This is a serious argument because it separates the label from the property. If the data say that certain ideas are “new” because they were published in prestigious contexts, a model can learn the context. If every novelty example is explained retrospectively, it can learn the narrative form of success. If unrecognized ideas are absent from the data, there is no direct feedback about them. In such a world, performance on a benchmark built from the same conventions may overestimate usefulness for discovering exceptions.

The argument is not dismissed by merely adding verbal reasoning. A system can produce an extensive justification of the same social label. Nor does a large training corpus automatically solve the problem: more examples of the same selection can reinforce the bias. Answering the skepticism requires evidence that the evaluator responds to controlled differences and can contradict a social cue when prior work refutes it.

A second skeptical objection concerns transformational novelty. A model trained to represent the world in a particular vocabulary may struggle when the contribution changes that vocabulary. Even if the architecture can express new combinations, the interpretation pro-

cedure may normalize the new framework to the old one. This objection does not prove impossibility, but identifies a region where standard tests built from stable categories may be insufficient.

A third objection concerns the lack of independent ground truth. If even experts cannot reliably establish novelty, how can we validate the detector? The answer cannot be a promise of omniscience. We can validate components in finite worlds, check concrete prior work and evaluate conditional decisions. For open conceptual changes, uncertainty and longitudinal follow-up must be accepted. The absence of a perfect standard does not mean every method is equally good; it means performance claims must be limited to what was measured.

Skepticism becomes unproductive when it changes its criterion after every success. If every recognized difference is retrospectively declared “not truly new,” the hypothesis can no longer be refuted. A scientific thesis must specify in advance what would count as recognition. We can require generalization to unseen families, correct witnesses and robustness to reformulation. But we cannot require the model to recognize something indefinable and then use the failure of definition as evidence against it.

13.2. The Optimistic Case: Old Procedures Can Detect New Objects

The strongest version of optimism is procedural. A system need not have memorized a contribution to establish that it differs from prior work. It can learn comparison operations, use new sources and delegate checks to tools. Novelty of the object does not require novelty of every cognitive operation used to evaluate it. A microscope built from known principles can observe an unknown specimen; the analogy illustrates possibility rather than equating an LLM with a calibrated measuring instrument.

Optimism has a strong formal counterargument to universal impossibility: in a finite world, an exact rule can decide relative novelty, and an LLM can participate in translating or controlling the procedure. If LLM use includes access to such tools, the class of possible sys-

tems is broader than an isolated generator. Excluding tools by definition may be legitimate for a model benchmark, but not for a conclusion about the usefulness of all LLM applications.

There is, however, an optimistic trap. The fact that the system can produce a witness does not guarantee that the witness is correct. The fact that it can cite a source does not guarantee that the source supports the relation. The fact that it can formalize a claim does not guarantee faithful formalization. A chain of approximate components does not become verified merely because the final component uses exact logic. If the translation is wrong, the verifier can impeccably certify a different problem.

The procedural argument must therefore be accompanied by contracts between stages. The model proposes an interpretation; the source allows it to be checked. The model proposes a witness; an executor tests it. The model proposes prior work; a comparator checks the passages. Where no automated verifier exists, uncertainty remains explicit. This discipline is less spectacular than the promise of an “autonomous researcher,” but offers a clear experimental path.

Another excessive optimism claims that scaling up models will inevitably solve the problem. The arguments presented provide no such guarantee. More capacity may improve some operations and reduce errors, but does not resolve missing information, incoherent criteria or irreversible pipeline collisions. A larger final model may use the original text better; it cannot reliably deduce inaccessible prior work or a difference completely removed from its input.

Likewise, sampling diversity is not identical to interpretation diversity. Raising temperature can produce more formulations and hypotheses, but they may remain within the same conceptual family. An exploration system must track operational differences between interpretations, not just lexical variation. If ten answers propose the same mechanism under different names, we do not have ten research directions.

The defensible optimistic position is therefore conditional: some components of novelty evaluation can be usefully automated, and their limits can be measured and reduced through source access, better

representations and verifiers. It does not follow that an unaudited linguistic score can replace scientific judgment. Technical possibility begins a validation obligation rather than ending it.

13.3. An Intermediate Position That Is Not a Comfortable Compromise

Supporting possibility and limits simultaneously does not mean avoiding a verdict. We can firmly reject the inference “learned from the past, therefore incapable of new comparisons.” We can firmly accept the limit “difference lost, therefore impossible to recover from the remaining representation alone.” We can regard the existence of familiarity bias in novelty evaluation as likely but still configuration-dependent. These positions do not contradict one another because they concern different claims.

The most important conceptual shift replaces the question about the model’s essence with the question about the observational contract. What differences must the system distinguish? What information does it receive? What transformations are allowed? What witnesses can be checked? What type of error would count as failure? This approach is closer to verifiable engineering than the dispute over whether the model “truly understands.”

Nevertheless, the problem of understanding does not disappear. It is partly operationalized. A system able to identify assumptions, produce distinguishing consequences, resist reformulations and revise its verdict when prior work appears displays competence relevant to evaluation. This does not demonstrate a complete theory of understanding. We have defined practically important, testable behaviors. This conceptual modesty is a methodological virtue, not an abandonment of deep questions.

Novelty may sometimes require changing the question, not just the answer. A detector fixed to one criterion can be correct within its framework yet insufficient for research. Advanced systems should therefore be able to signal that a contribution challenges the criterion or that two interpretations produce different evaluations. But signaling this should not automatically be rewarded as originality. It must be followed by clarification and testing of the consequences.

The most fruitful version of the initial intuition is thus a hypothesis about preserving differences under interpretation. A system educated by the past can be very good at making texts intelligible in the past's terms. Precisely this competence can become a vulnerability when the task is to notice an exception requiring those terms to expand. Failure is not inevitable. But we must not assume that a fluent interpretation preserved everything that mattered.

This argument can ground a distinct research program: evaluating fidelity to rare differences, not merely average fidelity to content. A summary can be evaluated by how well it preserves the questions separating a candidate from prior work. A retriever can be evaluated by its ability to find sources refuting or confirming the difference, not just thematic sources. An evaluator can be evaluated through witnesses and conditions, not merely agreement with global grades.

The book's verdict is not that we should trust or distrust LLMs as a homogeneous whole. Trust must be attached to operations and evidence. We can trust a correctly executed witness, place less trust in an unchecked translation and very little in a claim of historical absence without documented search. Such a distribution of trust is more demanding and useful than a global label about model intelligence.

Chapter 14. The Research Program and Criteria for Real Progress

14.1. A Minimal but Decisive Experimental Core

A research program need not begin by building a very large system. It should begin with an experiment capable of separating explanations. The proposed core is a finite corpus of contributions with exact labels, equivalent reformulations and minimal semantic differences. The model is tested with prior work absent, supplied and retrieved, while the full representation is compared with a compressed one. Each factor corresponds to a distinct mechanism.

An illustrative pilot could use 160 independent cases, equally distributed among eight construction families. For each case, four presentations, three prior-work access conditions and two representation paths would produce 3,840 evaluations for one model. Three repetitions of each evaluation would produce 11,520 calls. These numbers are a planning example, not a power calculation or cost recommendation. The 11,520 outputs do not represent 11,520 independent ideas.

The pilot first examines instrument integrity: labels are correct, variants equivalent, witnesses work and logs are complete. Only then do we estimate effect sizes and plan the confirmatory study. If the generator has artifacts or the parser loses information, a large benchmark may yield only a precise estimate of an incorrectly defined problem.

The primary outcome should be sensitivity to semantic change and invariance to irrelevant change in form. These must be measured together. A model that never changes its verdict is stable under paraphrase but does not detect differences. A model that always changes its verdict is sensitive but not discriminative. Competence requires change when the label changes and stability when it does not.

A hypothesis map can be maintained in a compact register.

Hypothesis	The Contrast That Can Support or Weaken It
Stylistic familiarity affects the verdict	The same contribution, controlled styles, identical prior work.
Compression loses relevant differences	Original, compressed representation and recovery through source access.
Retrieval is the dominant bottleneck	Automatically retrieved prior work versus supplied decisive prior work.
Evaluative frequency in training is a cause	Exposure intervention with controlled resources and ground truth.
The verdict uses the difference, not just rhetoric	Verified witnesses and reversals of the reference library.
The filter loses rare novel contributions in use	Probabilistic audit of rejections with a fixed selection budget.

Each hypothesis has its own level of generalization. An effect in a small, locally adapted model is evidence of a mechanism, not a direct estimate for every large model. Success on circuits is evidence of formal comparison, not evidence of evaluating cultural transformations. The register must prevent unjustified transfer of conclusions between levels.

14.2. From mechanism to architecture and utility

After identifying a mechanism, the next step is a remedial intervention. If the problem is the loss of quantifiers, we can add explicit extraction of them and access to passages. If the problem is thematic search, we can add difference-oriented queries. If the problem is calibration, we can separate the verdict from abstention and evaluate probabilities on an independent set. The remedy must be chosen according to the diagnosis, not the popularity of a technique.

Each remedy must be tested through ablation. We compare the complete system with a version without the component in question, keeping budgets as comparable as possible. An improvement may come

from additional cost, not the claimed mechanism. A comparison with a simple baseline is also useful, for example a full-text comparator or an explicit rule in the finite domain. A sophisticated system that does not outperform a modest baseline cannot justify its value through complexity.

Real progress may also consist in reducing unjustified confidence. If the new architecture better identifies cases where information is insufficient, raw accuracy may not increase much, but the risk of automated decisions may fall. However, the cost of abstentions must be measured. A system that escalates everything is not a useful evaluator, even if it no longer makes mistakes on its own.

After the laboratory, transfer to real documents must be gradual. First, short passages with validated differences. Then, complete papers with adjudicated prior-work packages. Then, end-to-end search over versioned corpora. Finally, prospective streams. Each stage adds a source of uncertainty. If performance drops, the structure of the stages allows the degradation to be located.

Utility must be evaluated in relation to human work. Does the system save search time? Does it produce comparisons that experts can verify faster? Does it reduce rediscoveries? Does it retain more rare contributions within the same budget? These questions are more concrete than “can AI evaluate research?” A system can be valuable as a prior-work assistant and unsuitable as a final filter. This boundary must appear in the conclusions and the interface.

Replication must include different model families and conditions of use, without assuming that a commercial name defines a stable configuration. Versions, data, tools and parameters must be documented. If a service changes, historical results remain valid for the configuration tested, but do not automatically describe the current configuration. A mature program treats evaluation as a continuous process, not permanent certification.

14.3. What would validate or invalidate the initial intuition

The universal form of the intuition would be contradicted by robust success on new objects, with exact ground truth, a fixed criterion and conditions that prevent mere retrieval of a memorized label. The model does not need to discover a fundamental law of nature to contradict a universal claim of impossibility. A nontrivial domain in which comparison generalizes correctly to new instances is enough.

The behavioral form would be supported if familiar style or innovation rhetoric systematically changes the verdict for the same content beyond a relevant threshold, and the effect replicates. It would be weakened locally if adequately powered tests show invariance to these manipulations and sensitivity to real semantic changes. No single experiment closes the question for all domains, but it can delineate a zone of confidence.

The causal form concerning training would require interventions on exposure. A convincing result would show that repeatedly associating a pattern with the label “new” subsequently leads to overrating that pattern, although ground-truth novelty is controlled. If the intervention has no relevant effect in well-tested regimes, the explanation must be revised. Perhaps the problem comes mainly from the prompt, the criterion or the representation, not the presumed frequency.

The informational form has a different status. For a demonstrated relevant collision, the impossibility of perfect classification from the remaining representation follows mathematically. What must be tested empirically is whether, and how often, real pipelines produce such collisions or approximate losses. The theorem provides no estimate of their frequency. The experiment must establish its practical relevance, not prove the theorem again.

The institutional form would require observations of the flow of ideas and feedback. If filters systematically reduce the diversity of verified differences and an audit discovers relevant contributions among rejections, there are reasons to change policy. But the impact must be separated from resource shortages, idea quality and other filters. A plausible story about homogenization does not replace measurement.

At the end of the research, we may reach a conclusion that is less dramatic but more valuable than the initial hypothesis: models can detect certain types of novelty, fail on others, and an important part of the failure comes from losing differences in representations and evaluating social or rhetorical proxies. This conclusion would lead directly to better architectures, clearer criteria and audit policies.

The final thesis is this. The past does not make recognizing the new impossible; it supplies the rules, comparisons and prejudices through which we examine it. The decisive question is whether these tools allow the difference to remain visible. A novelty evaluator becomes credible not when it speaks convincingly about originality, but when it can show what is different, compared with what, under which conditions, and through what evidence it would change its conclusion.

Appendix A. Detailed mathematical proofs and examples

A.1. From frequencies to decisions: three calculations that must not be confused

Consider a world with two classes, new and old. We want to understand the distribution of texts, learning a label and choosing an action separately. These operations are often merged into a single intuition about “what is probable,” although they answer different questions. The following examples are didactic constructions, with numbers chosen for clarity, not empirical findings about language models.

First, suppose a corpus contains nine hundred descriptions of idea A and one hundred of idea B. An ideal model of idea identity would assign probabilities of 0.9 and 0.1. If we always choose the most probable idea, we will produce only A. If we sample correctly from the distribution, we will also produce B, in its proportion. If we ask whether B differs from A and provide their definitions, the frequencies do not determine the answer on their own. We have moved from generation to comparison.

Now we introduce an evaluative label. In the data, descriptions with a certain rhetorical form S are called “innovative” in 80% of cases, and those without S in 20%. A classifier using only S can perform well if this correlation persists in the test. To find out whether it understands the conceptual difference, we construct a test in which S is independent of exact novelty. If performance falls, we have evidence that the cue mattered. To demonstrate that those frequencies caused its use, we must intervene on the data, not merely observe the correlation.

Finally, we choose an action. Even if the probability of novelty is estimated correctly, the decision depends on costs. If checking a candidate is cheap and losing a discovery is costly, the selection threshold can be low. If checking is very expensive, the threshold may rise. Two

systems with the same estimated probability can make different decisions without either being less capable of detecting novelty. The difference belongs to policy.

We now derive the decomposition of logarithmic loss. For a finite set and distributions P and Q , the average cost is the sum of $P(x)$ multiplied by minus $\log Q(x)$. We add and subtract minus $\log P(x)$. We obtain the entropy of P and the sum of $P(x) \log(P(x)/Q(x))$, namely the KL divergence. If Q assigns zero probability to an event with positive P , the cost is infinite. Otherwise, the divergence is nonnegative, and the ideal minimum is attained at $Q=P$.

$$-\sum_x P(x) \log Q(x) = -\sum_x P(x) \log P(x) + \sum_x P(x) \log \frac{P(x)}{Q(x)}.$$

Why is the divergence nonnegative? We can use the inequality $\log u \leq u-1$ for positive u . Applied to the ratio $Q(x)/P(x)$, it shows that the sum of $P(x) \log(Q(x)/P(x))$ is at most the sum of $Q(x)$ minus the sum of $P(x)$, namely zero. Changing the sign gives nonnegativity. This elementary proof explains why the ideal objective does not require eliminating rare events. It requires assigning correct probabilities, including to them.

In a model with limited capacity, the family Q may not contain P . Optimization then chooses the best approximation according to average loss. If errors on a rare class cost little in the overall average, the model may sacrifice that class for larger improvements in frequent regions. This is a possible tradeoff mechanism. It is not equivalent to the claim that every rare class will be ignored. A compact rule can simultaneously explain frequent and rare cases.

Returning to the prevalence example, for $\pi=0.001$, $\text{TPR}=0.9$ and $\text{FPR}=0.01$, the probability that a selection is truly new is 0.0009 divided by 0.01089, or approximately 0.08264. A detector with these characteristics recovers most novel contributions but produces many false alarms. If we lower FPR to 0.001, hypothetically maintaining the same sensitivity, precision becomes approximately 0.474. The example shows how much the false alarm rate matters in a rare-event stream.

This analysis does not automatically justify raising the threshold. The threshold may also reduce TPR. The complete curve and verification capacity must be analyzed. If we choose only the safest candidates, we

may eliminate precisely the unusual contributions. A selection policy can therefore retain a main channel with high precision and an audit or exploration channel. Their proportions must be set in relation to costs and evaluated, not chosen to produce a convenient score.

A.2. Bayes risk, collision and the possible benefit of a summary

Let us have a discrete representation Z and a binary label Y . For each value z , a classifier must choose zero or one. If it chooses zero, it errs on the joint mass $P(Y=1, Z=z)$. If it chooses one, it errs on the mass $P(Y=0, Z=z)$. The optimal choice takes the class with greater mass, and the local cost is the smaller mass. The sum of these minima gives the Bayes risk in Chapter 6.

This argument shows exactly where inevitable error arises. Not in stylistic complexity or insufficient reasoning, but in the fact that the same observation is compatible with different labels. A better algorithm can attain Bayes risk, but cannot improve on it without additional information or changing the problem. In a finite world with deterministic labels and complete observations, the risk can be zero. After compression, it can become positive.

For equal priors, write $p_0(z)=P(Z=z|Y=0)$ and $p_1(z)=P(Z=z|Y=1)$. The joint mass of each class is half its conditional distribution. We use the identity $\min(a,b)=(a+b-|a-b|)/2$. After summation, we obtain a risk equal to one half minus one quarter of the sum of absolute differences. Total variation is half that sum, which yields the formula in the chapter.

$$TV(p_0, p_1) = \frac{1}{2} \sum_z \text{abs}(p_0(z) - p_1(z)).$$

If p_0 and p_1 are identical, TV is zero and the minimum error is one half. If the distributions have disjoint supports, TV is one and the minimum error is zero. Between these extremes lie degrees of separability. In a compression experiment, we can ask whether the transformation makes the class distributions more similar with respect to useful information. We do not need to observe an exact collision of real-valued vectors to have practical degradation; they may become closer than the evaluator can resolve.

However, informational impossibility must be distinguished from difficulty for an algorithm. Suppose X is a long document that explicitly contains the correct label within a difficult logical relation. A summary Z expresses it clearly. The actual evaluator may have a lower error on Z . This does not contradict the fact that an optimal evaluator with access to X could compute the same summary and decide at least as well. The summary changes cost and accessibility; it does not create additional truth about the world.

A constructed example uses documents with many irrelevant sentences and two premises: if A then B ; A . The question is whether B follows. A faithful summary can retain only the premises and reduce errors by a limited model. But if the question is whether the document also asserts C , the same summary may be insufficient. Fidelity is task-dependent. A good summary for conclusion B may be poor for inventing all contributions.

We can formulate a compression objective oriented toward a family of questions Q . We want a low-cost representation such that answers to relevant questions are preserved with an error below a threshold. This is a design objective, not a complete method. The difficulty is that questions about novelty may be unknown before reading. If the family Q is too narrow, contributions that change the questions may be lost. If it is unlimited, the requirement approaches preserving all information.

A system can manage this tension through two levels. The compact representation serves frequent questions, while the original archive remains accessible for new questions. Cost shifts from complete storage in every context to selective access. The practical guarantee depends on the system's ability to recognize when it needs the source. Detecting insufficient representation is therefore a separate task that must be evaluated.

A subtle consequence is that the model's explanation may be more informative than its score and still insufficient for the verdict. If the explanation preserves a difference but the score ignores it, the information exists in the output and can be recovered by another aggregator. If the explanation and score were constructed after the same smoothing, both may miss the difference. Evaluation should inspect intermediate objects, not only the final label.

A.3. Rare differences and selection models: exact calculations

We construct two functions on eight bits. The first returns the XOR of the first two bits. The second returns the same value, except on the input where all eight bits are one, where the value is inverted. The functions differ on exactly one input out of 256. If the library contains only the first function, the second is new under the criterion of extensional equivalence. This label does not depend on how frequently the distinguishing input appears in use.

If inputs are uniform and we observe one hundred independent examples with replacement, the probability of not seeing the difference is $(255/256)$ to the power of one hundred, approximately 0.676. With one thousand examples, the probability is approximately 0.01996. To obtain a miss probability of at most 0.05, we need at least 766 observations in this regime. Enumerating all 256 inputs without replacement is, of course, sufficient. The sampling regime matters.

If the distinguishing input has probability ϵ under the real distribution, the general formula is $(1-\epsilon)$ to the power m . For small ϵ , the exponential approximation $\exp(-m\epsilon)$ shows that the required budget grows roughly inversely with ϵ . This is a reason not to equate the absence of an observed difference with functional equivalence. A test can be very long and still miss a rare case if it is not designed to seek it.

A symbolic verifier can sometimes avoid sampling by analyzing structure. For the functions above, it directly observes the condition that inverts the output. But we must not generalize from this simple construction to every program. Finding a witness may be computationally difficult. The difference between verifying a supplied witness and discovering one is central to the detector's operational complexity.

Now consider compression. If we retain the outputs on the first 255 inputs in lexicographic order and remove the last input, the two functions receive the same representation. The error comes from a deterministic projection, not a language model. If we later add the final row, the difference becomes identifiable. The example is an executable demonstration of how the verdict depends on the information retained.

For repeated selection, consider two classes with equal initial weights. The first is accepted with probability 0.55, the second with 0.50, and only accepted cases return. After t rounds, the weight ratio is 1.1 to the power t . After twenty rounds, the ratio is approximately 6.73. We have not modeled idea quality, the emergence of new contributions or error correction. The calculation shows only how a persistent asymmetry can be amplified in a closed process.

If we introduce external ideas in each round and maintain auditing, the dynamics change. Disadvantaged classes can be reintroduced, and acceptance probabilities can change after errors are discovered. Thus, the simple model is not an inevitable prediction of homogenization. It is a tool for formulating the variables to be measured: external input, selection probability, reevaluation rate and their dependence on familiarity.

These calculations share a purpose. They show that an observed error can have very different sources: the rarity of a distinguishing case, information compression or a repeated selection policy. None should automatically be attributed to the essence of an LLM. Instead, each can be introduced into a controlled experiment and lead to a concrete intervention.

Appendix B. Preregistration protocol and evidence register

B.1. Study record and inferential contract

The protocol below is a conceptual template for a confirmatory study. Concrete values must be set before data collection, following an independent pilot. It is not presented as an already registered protocol and does not imply that an experiment has been performed. Its purpose is to prevent retrospective changes to hypotheses, metrics and the population until the result becomes convenient.

The study title must name the mechanism and domain: for example, the influence of stylistic familiarity on detecting functional novelty in finite circuits. A formulation such as “can LLMs be creative?” is too broad for such an experiment. The target population must specify object families and description languages. Conclusions will not automatically extend to open-ended scientific ideas.

The primary hypothesis must be formulated as a measurable contrast. For example, the rate of correctly identifying the difference is changed by style when semantics and prior work are held constant. The direction can be preregistered, or the test can be two-sided if there are reasons for effects in both directions. A threshold for the smallest relevant effect must be chosen, expressed in percentage points, score difference or verification cost. The threshold must not be chosen after seeing the results.

Experimental conditions must define exactly what the model receives. An absent corpus, automated retrieval and a package of decisive prior work are different treatments. Full text, a summary and a summary with source access are other treatments. In a partial design, the reasons for omitting certain combinations must be declared. A reduced design can be very good if it answers a precise question.

The main independent unit is the contribution or family, not the number of generated responses. Repetitions estimate model variability in the same situation; they do not multiply the number of ideas. The

clustering structure and method for calculating intervals must be specified. If we use multiple models, we must declare whether inference concerns individual models or a family of training procedures.

Preregistered item	Required specification
Population	The domain, contribution families and inclusion criteria.
Ground truth	Exact verifier or expert procedure, with stated limits.
Manipulations	Semantics, style, access, compression and possibly exposure.
Primary outcome	A single metric or restricted family, chosen before the test.
Analysis	Contrasts, interactions, clustering and handling multiplicity.
Exclusions	Generator errors, invalid inputs and rules established beforehand.
Abstentions	Explicit treatment in coverage, risk and cost.
Practical relevance	The effect threshold that would change a decision about use.

Exclusions must concern real data defects, not inconvenient results. A case with an ambiguous label can be moved to a separate stratum under a prior rule. It must not be removed because the model answered differently from what the researcher expected. A response in the wrong format may be an instruction-following error and must be reported as such; it must not be secretly corrected by hand.

The analysis plan must distinguish confirmation from exploration. We can later investigate why a model fails on a family, but the result generates hypotheses. If we change the prompt after inspecting the test, we have a new procedure and need a separate evaluation set. A benchmark repeatedly used for tuning gradually becomes part of the system’s training, even if the model weights do not change.

B.2. A case register and the path from text to verdict

Each case must have a stable identifier and a version. The identifier must not reveal the label. Data must separate the semantic object from its presentation. For a circuit, the object can be the expression tree and truth table. The presentation can be natural-language text. For an article, the semantic object is not known exactly, so we retain the extracted contribution together with passages and alternative interpretations.

The register must include the novelty criterion and exact library. If the library changes, the label may change without modifying the candidate. This is not an inconsistency but a consequence of the relative definition. Versions must be retained so that the initial verdict can be reconstructed. In a temporal study, the reference time is part of the case, not merely decorative metadata.

Transformations must have provenance. Who or what produced the paraphrase? Which rules guarantee equivalence? Was it checked by an executor, an expert or only a model? For summaries, a record of what information was omitted must be retained. If evaluation later discovers a difference, we can determine whether it existed in the model's input or had been removed beforehand.

The evaluator's output must be stored in full, before any normalization. A parser can extract the score and verdict, but the raw output remains necessary for auditing. Parser errors must be tested separately. If the model says "I cannot decide" and the parser extracts a number from an example, the analysis may produce a false result without the model having issued that verdict.

Witnesses must be represented as verifiable objects. For a circuit, a witness is an input and the two outputs. For prior work, it is the source, version and passage supporting anticipation. For an empirical hypothesis, it may be a protocol that would produce different results under two mechanisms. A verified witness must be distinguished from a proposed witness. An idea about how the difference could be tested is not already proof of the difference.

B.3. Reporting the result and deciding whether to continue

The final report must present the primary result even if it does not support the hypothesis. It must show effect size, uncertainty, cost and analyses by preregistered families. Selecting a few spectacular examples is not enough. Examples are useful for the mechanism, but do not replace the distribution of results. A dramatic failure may be rare; a small bias may be systematic.

A null result must be interpreted in relation to the precision of the estimate. If the interval is wide, the study may be inconclusive. If the interval excludes effects greater than the practical threshold, we can say that evidence for a relevant effect has weakened in the configuration tested. Lack of significance must not be converted into proof of absence. Conversely, a significant difference does not automatically justify a verdict about operational importance.

The decision to continue should depend on the information gained. If the style effect is small but compression loss is large, the project can shift its focus toward representations. If retrieval is the main bottleneck, investment in a more expensive evaluator may be premature. If no remedy outperforms the simple baseline at comparable cost, the architecture must be simplified or the hypothesis revised.

The report must distinguish what the experimenter supplied, what was learned, what the model generated and what was verified exactly. This distinction is crucial for hybrid systems. Good performance may come from a perfect prior-work package or a generator that makes the task too easy. These advantages do not invalidate the experiment if declared, but they limit the conclusions.

The report's final section should state what future observation would change the conclusion. A new domain, another transformation family or an audit of rejections may reveal limits. A good scientific conclusion is not protected against revision; it is formulated so that revision is possible and informative.

Appendix C. An executable example of relative novelty

C.1. What the program verifies and what it does not

The program below is a complete Python example using only the standard library. It was executed to verify the calculations and finite properties presented here. It does not call an LLM, test a model's creativity or provide empirical evidence for familiarity bias. It independently verifies several conditions that a future language benchmark should satisfy.

A circuit is represented as a tree built from tuples. An integer denotes a variable's index. A tuple begins with the operator and continues with the arguments. Evaluation is recursive and does not dynamically execute code strings. The number of inputs is explicitly limited to avoid accidentally very large enumerations. The program's limit is a practical safeguard, not a theoretical limit of the method.

The first test shows that a reformulation using De Morgan's laws is equivalent to AND and therefore old relative to the chosen library. The second shows that XNOR is not in the library consisting of AND, OR and XOR. This novelty is relative; adding XNOR to the library would change the verdict. The program does not claim that XNOR is a new invention in the history of logic.

The third test constructs two functions on eight bits that differ only when all inputs are one. Verification enumerates all inputs and identifies the unique witness. It then removes the last row from the tables and confirms the collision of the remaining representations. The example thus separates a real difference from the information available to the evaluator. No classifier receiving only the two truncated tables can reliably deduce the missing difference without additional assumptions.

The final calculations use independent sampling with replacement and uniform inputs. They describe neither exhaustive enumeration nor natural input distributions. This qualification is necessary to in-

interpret the number of observations. The program can be saved to a file and run with a Python 3 interpreter; its deterministic output is reproduced after the code.

C.2. Complete implementation

```
"""Exemplu didactic; nu evalueaza niciun LLM."""
from itertools import product
from math import ceil, log

def value(node, bits):
    """Evalueaza un arbore validat, fara eval()."""
    if type(node) is int:
        if not 0 <= node < len(bits):
            raise ValueError("Indice de variabila invalid")
        return bits[node]
    if not isinstance(node, tuple) or not node:
        raise ValueError("Nod invalid")
    op, *args = node
    arity = {"not": 1, "and": 2, "or": 2, "xor": 2}
    if not isinstance(op, str) or op not in arity:
        raise ValueError("Operator invalid")
    if len(args) != arity[op]:
        raise ValueError("Operator sau aritate invalida")
    values = [value(arg, bits) for arg in args]
    if op == "not":
        return 1 - values[0]
    a, b = values
    if op == "and":
        return a & b
    if op == "or":
        return a | b
    return a ^ b

def table(node, n):
    if type(n) is not int or not 1 <= n <= 16:
        raise ValueError("n trebuie sa fie intre 1 si 16")
    return tuple(value(node, bits)
                  for bits in product((0, 1), repeat=n))

def compare(candidate, library, n):
    """Noutate functionala data de o biblioteca finita."""
    target = table(candidate, n)
    matches = [name for name, node in library.items()
               if table(node, n) == target]
    return not matches, matches

def all_inputs_true(n):
    node = 0
    for index in range(1, n):
```

```

        node = ("and", node, index)
    return node

def main():
    library = {"AND": ("and", 0, 1),
               "OR": ("or", 0, 1),
               "XOR": ("xor", 0, 1)}
    old = ("not", ("or", ("not", 0), ("not", 1)))
    new = ("not", ("xor", 0, 1))
    assert compare(old, library, 2) == (False, ["AND"])
    assert compare(new, library, 2) == (True, [])
    print("Parafraza AND: veche")
    print("XNOR fata de biblioteca: noua")

    n = 8
    baseline = ("xor", 0, 1)
    candidate = ("xor", baseline, all_inputs_true(n))
    first, second = table(baseline, n), table(candidate, n)
    witnesses = [bits for bits, a, b in
                  zip(product((0, 1), repeat=n),
                      first, second) if a != b]
    assert witnesses == [(1,) * n]
    assert first[:-1] == second[:-1]
    assert first != second
    print("Martor unic:", witnesses[0])
    print("Coliziune dupa eliminarea martorului: True")
    miss = (1 - 2 ** (-n)) ** 100
    needed = ceil(log(0.05) / log(1 - 2 ** (-n)))
    print(f"Ratare dupa 100 observatii: {miss:.6f}")
    print("Observatii pentru ratare <= 0.05:", needed)
    print("Toate verificarile exacte au trecut.")

if __name__ == "__main__":
    main()

```

C.3. Verified output and legitimate extensions

```

Parafraza AND: veche
XNOR fata de biblioteca: noua
Martor unic: (1, 1, 1, 1, 1, 1, 1, 1)
Coliziune dupa eliminarea martorului: True
Ratare dupa 100 observatii: 0.676116
Observatii pentru ratare <= 0.05: 766
Toate verificarile exacte au trecut.

```

These checks validate the didactic constructions, not the proposed language-based detector. Turning the example into a benchmark requires adding independent family generation, artifact balancing, language

variants, the evaluator interface and statistical analysis. The program's assertions should be retained in the generator so that no label is produced solely on the test author's intention.

Appendix D. Technical vocabulary and prerequisite knowledge

D.1. Probability, information and statistics

Random variable. A function that assigns a numerical or categorical outcome to each possible situation in a probabilistic model. In this book, the novelty label, evaluator verdict and document representation can be random variables. Randomness can describe variation across cases, not necessarily a fundamentally nondeterministic physical world.

Distribution. The rule assigning probabilities to a variable's values. The population it describes must be specified: training texts, ideas in a stream, circuit inputs or model outputs. An unannounced shift from one distribution to another is an important source of confusion in discussions of rarity.

Conditional probability. The probability of an event after fixing some information. An idea's overall frequency is not the same as the probability of the label "new" after comparing it with explicit prior work. Conditioning can radically change the distribution relevant to the decision.

Prior. The distribution used before observing the current information, relative to a probabilistic model. A prior is not necessarily an unjustified prejudice. It becomes problematic if it dominates relevant evidence or is confused with a verified conclusion about the individual case.

Posterior. The distribution updated after observing data, under the model's assumptions. A posterior can be calculated impeccably in a misspecified model. The resulting numerical confidence does not remove the obligation to verify that variables and assumptions match the novelty problem.

Likelihood. The value of the probability of the data viewed as a function of a model's parameters. It must not be confused with the probability that a scientific hypothesis is true or with an idea's novelty. A linguistically predictable text can contain an important conceptual difference.

Informational surprisal. The negative logarithm of an outcome's probability. It measures improbability relative to a distribution, not value, meaning or historical priority. Noise can be surprising, and an important contribution can be expressed very simply.

Entropy. A variable's average surprisal. It is low when the distribution is concentrated and high when many outcomes are difficult to anticipate. A corpus's entropy is not a direct measure of the originality of its ideas.

Cross-entropy. The average logarithmic cost of predictions from a distribution Q for data drawn from P . It decomposes into the entropy of P and the KL divergence. In training, minimizing it encourages prediction of the distribution, not merely selection of the most frequent outcome.

Kullback–Leibler divergence. An asymmetric measure of the difference between two distributions, relevant to prediction cost. It is not an ordinary geometric distance and does not automatically measure the semantic difference between two ideas. Choosing the variables is as important as the formula.

Mutual information. The amount by which observing one variable reduces uncertainty about another. In evaluating representations, we are interested in information about the relevant label or question, not only information about the document as a whole. A representation can retain the topic and lose the difference.

Total variation. A measure of separation between two distributions. In binary classification with equal priors, it determines the minimum Bayes risk. If the observation distributions are identical in the two classes, the observation cannot distinguish them better than the prior.

Bayes risk. The smallest possible expected error for the specified information and loss. It is not the error of a particular model. An algorithm can be far worse than the informational limit, and a representation can impose positive error even on an optimal algorithm.

Task-sufficient statistic. A representation that preserves everything needed for the intended decision. In the book's operational sense, sufficiency is relative to the criterion and questions. A representation sufficient for topic classification may be insufficient for mechanism novelty.

Distribution shift. The difference between the population used in training or calibration and the population encountered later. In novelty research, the shift is not merely an accidental malfunction; it may be the very phenomenon of interest. Guarantees obtained from the past must be interpreted conditionally.

Prevalence. The proportion of positive cases in the evaluated population. Low prevalence can produce many false alarms even for a detector with good sensitivity. Selection precision cannot be interpreted without knowing or estimating the relevant prevalence.

Precision. The proportion of positive selections that are correct. In an idea filter, it answers how much verification is spent on false alarms. It does not say how many new contributions were lost among rejections.

Recall or sensitivity. The proportion of truly positive cases detected. For discovery, it measures the ability to retain novel contributions. It can be estimated only if we also have information about rejected cases, not just selections.

False positive rate. The proportion of old cases classified as new. It differs from the proportion of false alarms among selections. Confusing these two denominators is one of the most frequent errors in interpreting rare-event detectors.

Calibration. Agreement between stated probabilities and observed frequencies of correctness. A system can be calibrated on average and poorly calibrated on a rare class. Calibration does not certify the fidelity of explanations or completeness of the prior-work corpus.

Brier score. The mean squared difference between predicted probabilities and observed binary labels. It measures the quality of probabilistic predictions. For novelty, labels must be defined beforehand; the formula does not resolve ambiguity in the criterion.

Selective classification. A procedure allowing abstention on some cases. Coverage, risk on decided cases and escalation cost must be evaluated together. Abstention must neither be automatically penalized as an error nor excluded in a way that makes performance look better.

Confidence interval. An interval produced by a procedure with a certain coverage property across repetitions of the study, under its assumptions. It does not automatically mean the probability that a fixed parameter lies in the particular observed interval. Interpretation requires attention to design and dependencies.

Effect size. The amount of difference between conditions, expressed in a relevant unit. Statistical significance alone does not say whether the effect matters for idea selection. A small effect can become important at large scale, but this must be analyzed explicitly.

Statistical power. The probability that a test detects a specified effect under a model of the data. It is not a universal property of a number of examples. It depends on effect size, variation, correlations and the analysis procedure.

Clustering and pseudoreplication. Observations from the same idea or family may be dependent. Treating all paraphrases and repetitions as independent can underestimate uncertainty. Analysis must reflect the level at which real replications exist.

D.2. Language models, representations and retrieval

Token. The discrete unit processed by a language model. It can be a word, part of a word or a symbol. Comparing length and perplexity across different tokenizations requires caution because the units do not have the same granularity.

Embedding. A vector representation of an object, token or passage. Vector proximity can be useful for retrieval, but is not identical to semantic equivalence. The training objective determines which similarities are favored and which differences may be ignored.

Attention. A mechanism for combining information across positions in Transformer architectures. It enables context-dependent access to other representations. By itself, it does not guarantee that the model uses all premises relevant to a judgment.

Autoregressive. A mode of factorization and generation in which each element is conditioned on preceding elements. It describes the probabilistic organization of the output, not the full range of operations that can be performed in computing that output.

Pretraining. A stage in which the model learns from a broad corpus, usually through a general objective. Hypotheses about corpus frequencies must be separated from the effects of subsequent tuning, the prompt and the tools available during use.

Post-training. Stages that modify model behavior after initial learning, for example through demonstrations or preferences. They can change style and instruction following. An effect observed in the final model cannot automatically be attributed solely to pretraining.

SFT. Supervised fine-tuning on examples of inputs and desired outputs. For novelty evaluation, example selection and labeling criteria matter: the model may learn a comparison procedure or cues to the desired verdicts.

RLHF. A family of learning procedures using human feedback that involve optimizing a policy. The reward used is a proxy for the intended preferences or criteria. It must not be assumed that the reward coincides with the historical truth of a contribution.

DPO. Direct preference optimization through an objective based on pairs of preferred and dispreferred responses. What matters in this book is that preferences can change the final policy; we do not attribute a universal effect on novelty to the method.

In-context learning. Changing behavior through examples and instructions supplied in the prompt, without updating parameters during that interaction. It allows testing local worlds with new symbols, but does not guarantee learning every rule from a few examples.

Decoding. The procedure that turns model distributions into a concrete output. Greedy selection, sampling and other strategies can produce different results. Changing decoding must not be confused with changing competence in verifying prior work.

Temperature. A parameter that changes the concentration of the distribution used for sampling. Higher temperature can increase textual variation. It does not guarantee conceptual diversity, correctness or historical originality.

Perplexity. The exponential of average logarithmic loss over the chosen units. It is a text-prediction metric, dependent on the model, context and tokenization. It can be a cue to familiarity, but not a direct measure of novelty.

LLM-as-a-judge. Using a language model to evaluate other outputs or documents. The criterion, format and sources must be specified. Agreement with human evaluators on one task does not automatically transfer to other types of judgment.

Familiarity bias. Sensitivity of evaluation to a match with familiar patterns when this is not justified by the task criterion. Demonstrating it requires controlling variables such as clarity and correctness, not merely observing an association with style.

Learning shortcut. Using a cue correlated with the label instead of the intended relation. It can produce good results on the usual distribution and failures when the cue is decoupled from truth. Factorial experiments attempt to create precisely this decoupling.

Semantic smoothing. A working term for transformations that reduce variation between representations. It is useful when it removes irrelevant differences and dangerous when it erases decisive assumptions, exceptions or relations. It is not treated as an inevitably negative property.

RAG. Generation augmented by document retrieval. It adds information accessible at evaluation time, but does not guarantee corpus completeness or correct comparison. Retrieval and judgment must be measured separately.

Lexical retrieval. Search based on terms and surface matches. It can be valuable for formulas, names and rare expressions. It may miss reformulated prior work, but should not be discarded in favor of a semantic representation that loses exact details.

Semantic retrieval. Search based on representations of meaning or context. It helps overcome vocabulary differences. Retrieved similarity is a lead for comparison, not proof that prior work anticipates the contribution.

Cross-encoder. A model that processes two texts together to evaluate their relationship. It can use detailed interactions, but has different costs from independent indexing. The choice of technique must be evaluated at the same operational budget.

Late interaction. A strategy in which representations of text parts are calculated separately and later combined for scoring. It can preserve more granularity than a single global vector. However, it does not automatically certify preservation of novelty differences.

Top-k. The first k results according to a retrieval or evaluation score. Decisive prior work may be absent from this subset even if it exists in the index. Choosing k is a cost and coverage decision, not a guarantee of completeness.

Contamination. A situation in which information from the test or about its answer was available during preparation of the model or procedure. It can also arise through repeated prompt tuning on the test. Superficial anonymization does not automatically eliminate it.

D.3. Semantics, circuits, compilers and verification

Syntax. The form of an expression according to a language's rules. Two syntactically different texts can have the same behavior. A novelty detector that measures only changes in form will confuse paraphrases with contributions.

Semantics. The rule or family of rules through which expressions receive an interpretation. For circuits, it can be the computed function; for a proposal, it can include assumptions and guarantees. Not all novelty questions are required to use the same semantics.

Extensional semantics. Describing a program through the relation between inputs and outputs. In its simple form, it ignores cost, internal structure and some observable effects. It is exact for certain questions and insufficient for others.

Observational equivalence. The indistinguishability of two systems relative to a declared class of observations. If we change the permitted observations, for example by adding time or disclosed information, equivalence may disappear. The observational criterion must be declared.

AST. An abstract syntax tree, representing an expression's structure without all graphical details. Tuples play this role in the executable example. The AST enables evaluation and transformations, but does not by itself guarantee fidelity of translation from natural language.

Intermediate representation. An internal form used between reading the source and execution or analysis. It can facilitate comparison and verification. If it loses a relevant difference, later stages cannot recover it from that representation alone.

Compiler. A procedure that translates between representations with the aim of preserving specified properties. The analogy with extracting ideas is useful: what is preserved must be declared. A fluent translation is not automatically semantically correct.

SSA. A representation form in which each variable is assigned only once, useful for tracing dependencies. In a contribution-analysis graph, a similar discipline can help version results. It does not, by itself, confer truth on stored claims.

Boolean circuit. A finite graph of logical operations on binary inputs. It has exact semantics over a finite domain and enables exhaustive verification at small sizes. It is a controlled laboratory, not a complete model of scientific creativity.

Truth table. Enumeration of a Boolean function’s outputs for all inputs. Two circuits with the same table are extensionally equivalent on that domain. The number of rows grows exponentially with the number of inputs.

Witness. A concrete object supporting an existential claim, for example an input on which two functions differ. A witness proposed by the model must be verified. The existence of a correct witness can be more informative than an overall originality score.

Counterexample. A case that refutes a universal claim. If a system claims two programs are equivalent, a single input with different outputs is enough to refute it. The absence of a discovered counterexample does not automatically prove equivalence.

SAT. The problem of whether an assignment exists that makes a Boolean formula true. It can be used to search for a witness of difference between circuits. Using a solver requires a correct translation of the problem; the solver does not verify the text author’s intention.

SMT. Satisfiability in the presence of theories such as arithmetic or arrays. It extends the types of claims that can be formalized. A proof within a formalization does not guarantee that the formalization fully matches an informal proposal.

Miter. A construction that combines two circuits so that its output indicates a difference between them. Searching for an input that activates the output produces a witness of nonequivalence. If complete verification proves this impossible, we have equivalence in the chosen formal model.

Soundness, or logical correctness of conclusions. The property of a procedure not certifying false claims within its formal model. It must be separated from the model’s fidelity to the text or world being analyzed. A sound verifier can receive incorrect premises.

Completeness. A procedure’s ability to identify all true cases in the declared class. A system can be reliable when it certifies, yet abstain on many cases. It must not be confused with coverage of a document corpus.

Decidability. The existence of an algorithm that halts and answers correctly for every instance in a class. It differs from practical feasibility. A procedure can be decidable and too costly for large-scale use.

Undecidability. The absence of a total correct algorithm for all instances of a class. It does not mean that no instance can be solved or that approximate methods are useless. The limit must be related to the exact class of problems.

Complexity. The study of computational resources needed as a function of input size. For novelty, the cost of searching for prior work, translation and verification may dominate in different ways. An accuracy score without cost does not fully describe utility.

Datalog. A language of logical rules used for relations and inference, often over finite databases. It can represent dependencies or consequences in a controlled laboratory. The choice of rules and negation semantics must be explicit; it is not a universal novelty detector.

Logical closure. The set of consequences deducible from a base of claims under specified rules. If we include all ideal consequences, a new proof may not add a proposition that was not already implied. Epistemic progress may concern the consequence's effective accessibility.

Abstraction. Controlled removal of details to facilitate analysis. It is useful only relative to the properties it must preserve. In novelty evaluation, abstraction must be audited for loss of exceptions, assumptions and resource differences.

Provenance. Information about the origin and transformations of an object or verdict. It includes the source, version and dependencies. It does not prove truth, but allows conclusions to be verified, reproduced and revised.

D.4. Operational epistemology and experimental design

Relative novelty. The absence of relevant prior work in a declared reference corpus under a fixed criterion. It is the form of novelty most easily made testable. It must not automatically be extended to absolute historical priority.

Epistemic novelty. A change in what an agent or community knows or can justify. A new proof can play this role even if the result was implicit in the axioms. Resources and effective access matter.

Transformational novelty. A change to the rules, representation or problem space, not merely an additional solution within the existing framework. Identifying it requires examining the consequences of the framework change, not automatically rewarding unusual vocabulary.

Anticipation. The relation through which prior work already contains the relevant elements of the contribution being evaluated. It can be asymmetric and broader than textual identity. In this book it is an operational concept, not a legal definition.

Residual difference. What remains unanticipated after explicit comparison with prior work. It may be important, minor or uncertain. The term requires the evaluator to describe the difference instead of assigning only an impression-based score.

Construct. The theoretical property we want to measure, such as mechanism novelty. Metrics are operational proxies for the construct. Agreement with reviewers may measure community judgment without fully coinciding with historical priority.

Construct validity. The degree to which a procedure measures the intended property rather than other variables. A benchmark may be reliable yet primarily measure innovative style. Measurement consistency is not enough.

Identifiability. The possibility of determining the intended property from available observations and assumptions. Two worlds with identical observations and different historical verdicts make the property unidentifiable without additional information.

Causal confounding. A situation in which an association between two variables is influenced by other causes. For example, perplexity and score may be linked through clarity. A correlation alone does not identify training frequency as the mechanism.

Intervention. A controlled change to a variable, used to study its effect. Rewording text intervenes on presentation; retraining with controlled frequencies intervenes on exposure. The two do not answer the same causal question.

Factorial design. Systematically combining levels of multiple factors. It allows study of effects and interactions, for example whether style matters only when prior work is absent. The number of conditions must be balanced against budget and desired precision.

Interaction. A situation in which the effect of one factor depends on another. An overall average can hide opposing effects under different conditions. Important contrasts must therefore be defined beforehand, not selectively discovered after the results.

Experimental oracle. A condition in which the experimenter supplies the decisive information for comparison. It is not a claim of omniscience. Its role is to separate access to prior work from the ability to compare it.

Ablation. Removing a component to estimate its contribution to the system. Total cost must be controlled as far as possible. A component may appear useful only because it adds time or extra calls.

Preregistration. Fixing hypotheses, metrics and analysis rules before the test. It does not forbid exploration, but separates it from confirmation. It reduces the freedom to retrospectively choose only procedures favorable to the desired conclusion.

Audit of rejections. Checking a sample of cases eliminated by the filter. It is necessary to estimate missed novel contributions. Without it, we mainly observe false alarms among selections and may overestimate the system's utility.

Prospective evaluation. Testing on contributions collected after the procedure is fixed, before a stable public consensus. It reduces some forms of contamination but requires follow-up over time and acceptance of labels that remain undetermined.

Local falsifiability. The existence of observations that would weaken a precise claim about a model, representation or domain. It is more useful than a global thesis protected by continually redefining what “truly new” means.

Correctability. The system’s ability to revise a verdict when new sources, interpretations or witnesses appear. It is an architectural and epistemic property. A credible evaluator must preserve the path through which its conclusion can be refuted.

References and reading map

Primary sources consulted

The references below document claims about the existing literature. The working definitions, elementary proofs and protocols of this monograph must be distinguished from results reported in the sources. Metadata and versions were checked through September 11, 2026. For papers distributed through arXiv, that accessible version is indicated, without assuming that all results have been independently replicated. When the consulted page indicates a conference, that status is explicitly mentioned.

[R01] Boden, Margaret A. (2009). Computer Models of Creativity. *AI Magazine*, 30(3), 23. DOI: 10.1609/aimag.v30i3.2254. Publication page. Relevance: types of creativity and the difference between combination, exploration and transformation of a framework.

[R02] Shannon, Claude E. (1948). A Mathematical Theory of Communication. *The Bell System Technical Journal*, 27, 379–423 and 623–656. Article text, academic reproduction. Relevance: probabilistic information, entropy and separating the communication problem from the meaning of messages.

[R03] Vaswani, Ashish, and colleagues (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*. arXiv:1706.03762. Relevance: the Transformer architecture; it does not constitute proof of competence in novelty evaluation.

[R04] Brown, Tom B., and colleagues (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*. arXiv:2005.14165. Relevance: behaviors conditioned on examples in context and the distinction between instance memorization and task adaptation.

[R05] Ouyang, Long, and colleagues (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*. arXiv:2203.02155. Relevance: adjusting behavior through demonstrations and preferences; separating pretraining from the final policy.

[R06] Rafailov, Rafael; Sharma, Archit; Mitchell, Eric; Ermon, Stefano; Manning, Christopher D.; Finn, Chelsea (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *Advances in Neural Information Processing Systems*. arXiv:2305.18290. Relevance: direct preference optimization as a mechanism distinct from the initial language-modeling objective.

[R07] Wataoka, Koki; Takahashi, Tsubasa; Ri, Ryokan (2024). Self-Preference Bias in LLM-as-a-Judge. arXiv:2410.21819, version 1. Relevance: the relationship between familiarity to the model, perplexity and evaluator preferences. The study’s domain is not identical to historical scientific novelty.

[R08] Zheng, Lianmin, and colleagues (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS, Datasets and Benchmarks*. arXiv:2306.05685. Relevance: the utility of LLM evaluators and biases such as position, verbosity and self-preference.

[R09] Si, Chenglei; Yang, Diyi; Hashimoto, Tatsunori (2024). Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers. arXiv:2409.04109, version 1. Relevance: blind human evaluation of generated ideas; the difference between perceived novelty, feasibility and verified priority.

[R10] Gupta, Tarun; Pruthi, Danish (2025). All That Glitters is Not Novel: Plagiarism in AI Generated Research. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics. ACL Anthology*. Relevance: explicit examination of prior work for AI-generated research documents.

[R11] Schopf, Tim; Färber, Michael (2026). Is this Idea Novel? An Automated Benchmark for Judgment of Research Ideas. *RINoBench*. arXiv:2603.10303, version 1. The consulted metadata indicate acceptance at LREC 2026. Relevance: evaluation of novelty scores and justifications; construction and limitations of a reference standard derived from reviews.

[R12] Wu, Wenqing; Zhao, Yi; Wang, Yuzhuo; Li, Siyou; Shao, Juexi; Long, Yunfei; Zhang, Chengzhi (2026). NovBench: Evaluating Large Language Models on Academic Paper Novelty Assessment. arXiv:2604.11543, version 1. The consulted metadata indicate ACL 2026. Relevance: novelty evaluations based on introductions and reviews, with separate quality dimensions.

[R13] Liu, Yan; Yang, Zonglin; Poria, Soujanya; Nguyen, Thanh-Son; Cambria, Erik (2025). Harnessing Large Language Models for Scientific Novelty Detection. arXiv:2505.24615, version 1. Relevance: conceptual retrieval of prior work and source-augmented evaluation in novelty benchmarks.

[R14] Lewis, Patrick, and colleagues (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems. arXiv:2005.11401. Relevance: combining parametric memory with documents retrieved during use.

[R15] Khattab, Omar; Zaharia, Matei (2020). ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. SIGIR. arXiv:2004.12832. Relevance: retrieval with granular representations and late interaction; an access technique, not a certificate of equivalence.

[R16] Liu, Nelson F.; Lin, Kevin; Hewitt, John; Paranjape, Ashwin; Bevilacqua, Michele; Petroni, Fabio; Liang, Percy (2024). Lost in the Middle: How Language Models Use Long Contexts. Transactions of the Association for Computational Linguistics. arXiv:2307.03172. Relevance: performance dependence on information position in the contexts and models tested.

[R17] Nalisnick, Eric; Matsukawa, Akihiro; Teh, Yee Whye; Görür, Dilan; Lakshminarayanan, Balaji (2019). Do Deep Generative Models Know What They Don't Know? ICLR. arXiv:1810.09136. Relevance: limitations of using likelihood as a cue to distribution membership.

[R18] Guo, Chuan; Pleiss, Geoff; Sun, Yu; Weinberger, Kilian Q. (2017). On Calibration of Modern Neural Networks. Proceedings of Machine Learning Research, 70, 1321–1330. PMLR page. Relevance: separating numerical confidence from empirical correctness, and calibration procedures.

[R19] Saito, Takaya; Rehmsmeier, Marc (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. PLOS ONE, 10(3), e0118432. DOI:10.1371/journal.pone.0118432. Relevance: measuring detectors in imbalanced populations.

[R20] Angelopoulos, Anastasios N.; Bates, Stephen (2021; revised 2022). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv:2107.07511. Relevance: prediction with marginal guarantees and the importance of data assumptions; not certification of individual novelty.

[R21] Ren, Jie; Liu, Peter J.; Fertig, Emily; Snoek, Jasper; Poplin, Ryan; DePristo, Mark A.; Dillon, Joshua V.; Lakshminarayanan, Balaji (2019). Likelihood Ratios for Out-of-Distribution Detection. Advances in Neural Information Processing Systems. arXiv:1906.02845. Relevance: controlling the effects of background statistics in out-of-distribution detection.

[R22] Uzzi, Brian; Mukherjee, Satyam; Stringer, Michael; Jones, Ben (2013). Atypical Combinations and Scientific Impact. Science, 342(6157), 468–472. DOI:10.1126/science.1240474. Bibliographic record and abstract. Relevance: conventional and atypical combinations in scientific literature; citation impact is not identical to truth.

[R23] Shumailov, Ilia; Shumaylov, Zakhar; Zhao, Yiren; Papernot, Nicolas; Anderson, Ross; Gal, Yarin (2024). AI models collapse when trained on recursively generated data. Nature, 631, 755–759. DOI:10.1038/s41586-024-07566-y. Relevance: effects of certain recursive training regimes; not proof of inevitable collapse for every use of synthetic data.

[R24] Gerstgrasser, Matthias, and colleagues (2024). Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data. arXiv:2404.01413. Relevance: the importance of the difference between data accumulation and replacement in recursive regimes.

[R25] Hemaspaandra, Lane A.; Rothe, Jörg (1999). A Second Step Towards Complexity-Theoretic Analogs of Rice's Theorem. arXiv:cs/9907038. Relevance: formal context for semantic properties, undecidability and restricted circuit classes. It is not an empirical study of language models.

How the sources connect to the book's argument

For foundations in probability and architecture, the entry point is R02–R06. For the familiarity hypothesis and limits of automated evaluation, the core is R07–R12. R13–R16 are relevant to building a system with explicit access to prior work. For measurement and uncertainty, R17–R21 offer cautions and tools to adapt, not transfer mechanically. For creativity, combinations and data dynamics, R01 and R22–R24 complete the perspective. R25 delineates a type of formal impossibility that must not be confused with the practical difficulty of all tasks.